CHAPTER

Two

Magnetic Effects of the Electric Current and Measuring Instruments

Lesson 1 Magnetic Effect of the Electric Current.

Lesson 2 Following of the Magnetic Effect of the Electric Current.

Lesson 3 The Magnetic Force and the Torque.

Lesson 4 The Measuring Instruments.

- Questions on each lesson from the Previous Years Exams.
- Test on Chapter 2



CamScanner

Scanned with OKEN Scanner

CamScanner

Chapter 2

Questions on Lesson One

Magnetic Effect of the Electric Current

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

Magnetic flux (ϕ_m)

- If the field is:
  - Perpendicular to the area → $\phi_m = BA$
  - Making angle (θ) with the normal to the area → $\phi_m = BA \cos \theta$
  - Parallel to the area → $\phi_m = 0$
- If the coil is rotated with angle (θ) from:
  - Perpendicular position → $\phi_m = BA \cos \theta$
  - Parallel position → $\phi_m = BA \cos (90 - \theta)$

- **The magnetic flux density (B) at a point at a distance d from the axis of a straight wire carrying electric current of intensity I:** $B = \mu \frac{I}{2 \pi d}$
  Where: μ is the permeability of the medium

Scanned with OKEN Scanner

- **The magnetic flux density or the magnetic field (B) is a vector quantity, so if the magnetic flux density due to an electric current in a wire is B_1 at a certain point and the wire is placed in an external magnetic field of flux density B_2 then when:**
  - **The two fields have the same direction:** $B_t = B_1 + B_2$
  - **The two fields have opposite directions:** $B_t = B_1 - B_2$ $(B_1 > B_2)$
  - **The two fields are perpendicular to each other:** $B_t = \sqrt{B_1^2 + B_2^2}$

The net magnetic flux density due to the flow of electric currents in two parallel wires

If the two currents are in the same direction

- Outside the region between the two wires: $B_t = B_1 + B_2$
- Neutral point: At a point in the region between the two wires $B_1 = B_2$

$$\frac{\mu I_1}{2\pi(x-d)} = \frac{\mu I_2}{2\pi d}$$

$$\therefore \frac{I_1}{x-d} = \frac{I_2}{d}$$

Where: x is the distance between the two wires, d is the distance between the neutral point and the wire of the less current $(I_2 < I_1)$

- In the region between the two wires: $B_t = B_1 - B_2$ $(B_1 > B_2)$

If the two currents are in opposite directions

- Outside the region between the two wires: $B_t = B_1 - B_2$ $(B_1 > B_2)$
- Neutral point: At a point outside the region between the two wires $B_1 = B_2$

$$\frac{\mu I_1}{2\pi(x+d)} = \frac{\mu I_2}{2\pi d} \therefore \frac{I_1}{x+d} = \frac{I_2}{d}$$

Where: x is the distance between the two wires, d is the distance between the neutral point and the wire of less current $(I_2 < I_1)$

- In the region between the two wires: $B_t = B_1 + B_2$

CamScanner

Scanned with OKEN Scanner

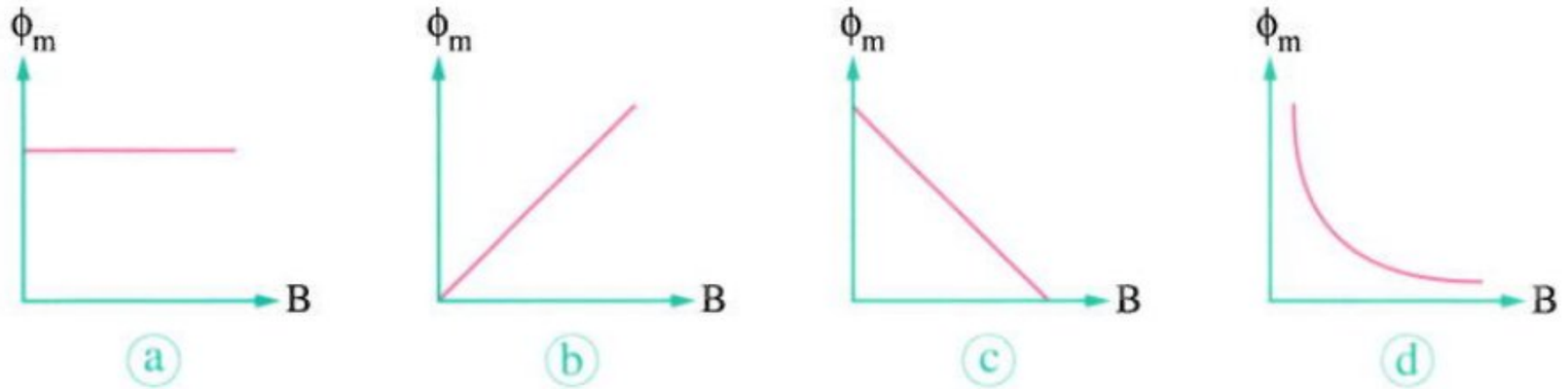
CamScanner

First Multiple choice questions

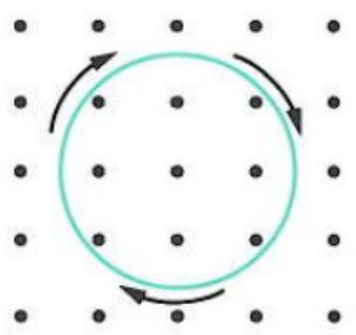
Use the following constants, if needed: $\mu_{air} = 4\pi \times 10^{-7}$ Wb/A.m, e = 1.6×10^{-19} C

Magnetic flux and flux density

1 A rectangular coil is placed perpendicular to a magnetic field of a uniformly changing strength. Which of the following graphs represents the relation between the total flux (ϕ_m) and the magnitude of the magnetic flux density (B) in which the coil exists?



2 The opposite figure represents a circular coil placed perpendicular to a uniform magnetic field. If the coil is rotated about its axis which is perpendicular to its plane in the clockwise direction by an angle of 90°, then the flux that penetrates the coil ..... .



(a) increases (b) equals zero

(c) decreases (d) does not change

3 * A ring of cross-sectional area 0.2 m^2 is placed perpendicular to a uniform magnetic flux of density 0.04 Wb/m^2. So, the magnetic flux which passes through this ring equals ..... .

(a) 0.002 Wb (b) 0.004 Wb (c) 0.006 Wb (d) 0.008 Wb

4 * A square frame of side length 20 cm is placed in a uniform magnetic field of flux density 3×10^{-2} tesla, if the flux which passes through the frame is 6×10^{-4} weber, the angle between the frame plane and the magnetic lines equals ..... .

(a) 20° (b) 30° (c) 45° (d) 90°

5 * A coil of cross-sectional area 2 m^2 is placed in a magnetic field of flux density 0.05 Wb/m^2 where the magnetic flux that passes through the coil is the maximum value, so the magnetic flux when the coil rotates with an angle of:

(i) 30°, equals ..... .

(a) 0.01 Wb (b) – 0.1 Wb (c) 0.7 Wb (d) 0.087 Wb

Scanned with OKEN Scanner

(ii) 45°, equals ..... .

(a) – 0.01 Wb (b) 0.03 Wb (c) 0.07 Wb (d) 0.09 Wb

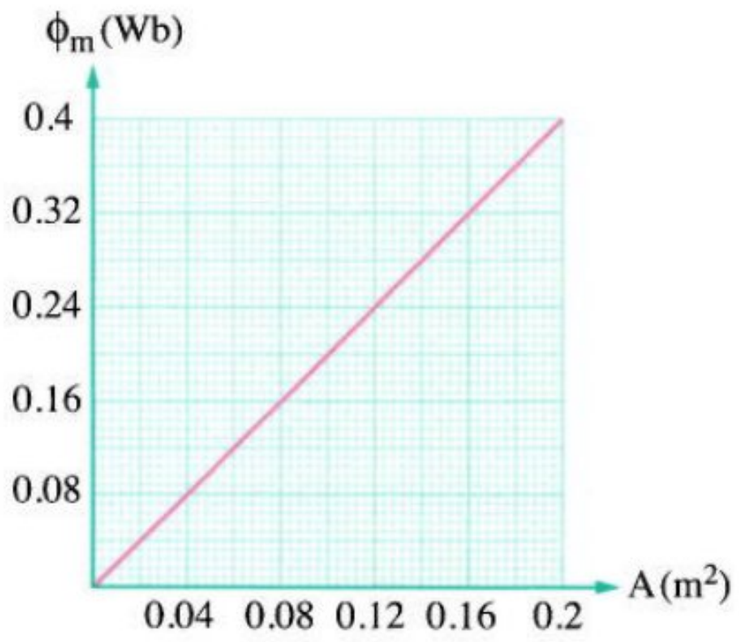
(iii) 135°, equals ..... .

(a) 0.1 Wb (b) 0.005 Wb (c) – 0.07 Wb (d) – 0.09 Wb

(iv) 180°, equals ..... .

(a) – 0.1 Wb (b) 0.2 Wb (c) – 0.3 Wb (d) 0.4 Wb

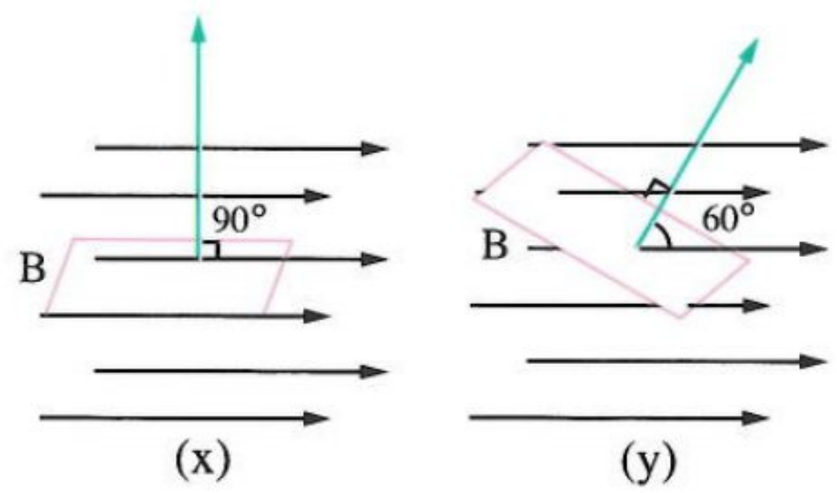
6 Multiple rectangular coils of different areas are placed in a uniform magnetic field where the coil planes have an angle 30° with the direction of the field and the opposite graph shows the relation between the total magnetic flux that passes through the coil (ϕ_m) and the area of the coil (A), so the magnetic flux density that affects all the coils is ..... .



(a) 1 T (b) 4 T

(c) 0.5 T (d) 0.8 T

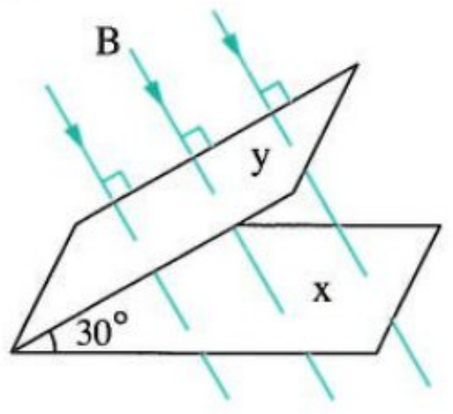
7 ✱ The opposite figure shows two different positions (x) and (y) for a coil of area 0.2 m^2 in a uniform magnetic field of flux density 0.8 T, so the change in the magnetic flux $\Delta\phi_m$ through the coil between the two positions equals ..... .



(a) 0 (b) 0.08 Wb

(c) 0.4 Wb (d) 0.16 Wb

8 Two rectangular coils of the same area are fixed together such that one is inclined relative to the other at an angle of 30° If the two coils are exposed to a magnetic flux of density B and its direction as shown in the opposite figure, then the ratio between the values of magnetic flux that penetrates the two coils $\left(\frac{(\phi_m)_x}{(\phi_m)_y}\right)$ equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2}{\sqrt{3}}$

CamScanner
Scanned with OKEN Scanner

CamScanner

9 A coil of face area A is placed perpendicular to a uniform magnetic field of flux density B, the magnitude of the change in magnetic flux that penetrates the coil when it rotates from the perpendicular position around an axis in the same plane:

(i) $\frac{1}{4}$ cycle equals ..... .

(a) zero (b) BA (c) $\frac{3}{2}$ BA (d) 2 BA

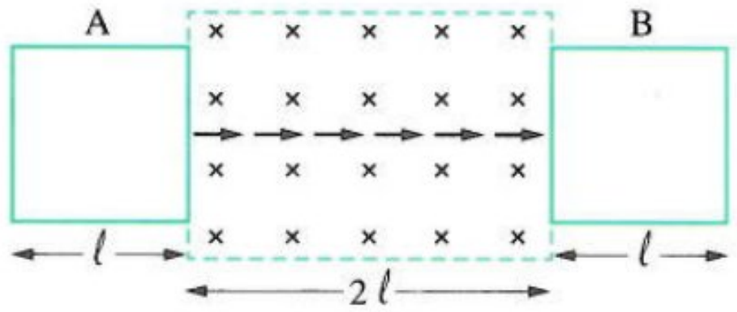
(ii) $\frac{1}{2}$ cycle equals ..... .

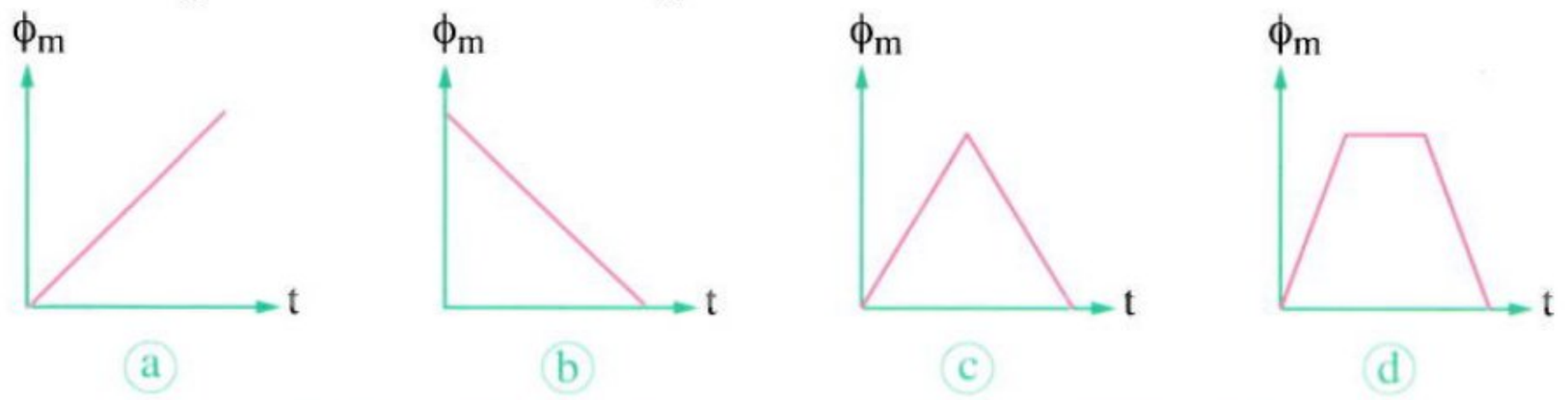
(a) zero (b) BA (c) $\frac{3}{2}$ BA (d) 2 BA

(iii) $\frac{1}{3}$ cycle equals ..... .

(a) zero (b) BA (c) $\frac{3}{2}$ BA (d) 2 BA

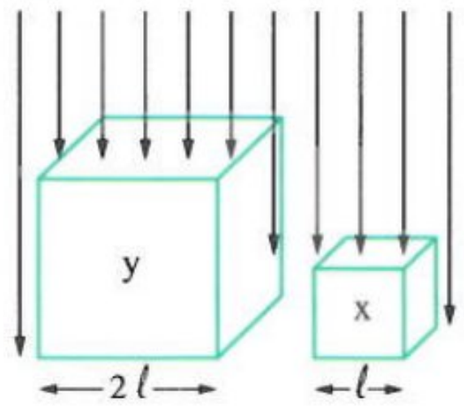
10 The opposite diagram shows a square coil that is moving with constant velocity towards the right direction of the page penetrating a uniform magnetic field which is normal into the page, so the relation between the magnetic flux (ϕ_m) that passes through the coil during its motion from position A to position B and the time (t) is ..... .





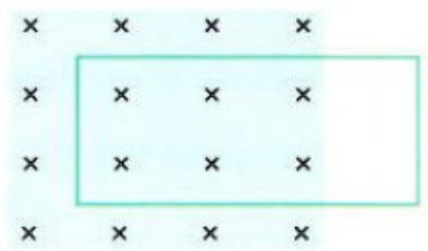
(a) (b) (c) (d)

11 Two cubes x and y that have side lengths ℓ and 2ℓ respectively are affected by a uniform magnetic field perpendicularly on the upper surface of each of them as shown in the opposite figure, so which of the following relationships represents the relation between the magnetic fluxes that pass through the upper surfaces of them?



(a) $(\phi_m)_y = \frac{1}{4}(\phi_m)_x$ (b) $(\phi_m)_y = (\phi_m)_x$ (c) $(\phi_m)_y = 2(\phi_m)_x$ (d) $(\phi_m)_y = 4(\phi_m)_x$

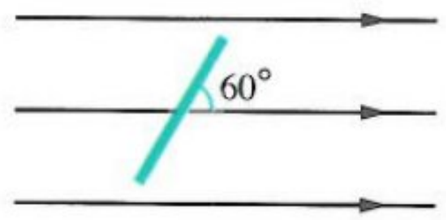
12 A coil of area A is placed perpendicular to a uniform magnetic field and a magnetic flux of 10^{-3} Wb passes through its face, when the coil is shifted such that the area through which the flux passes is decreased by 10 cm^2 as shown in the opposite figure, if the magnetic flux that passes through the coil becomes 7.5×10^{-4} Wb, then the area of the coil (A) equals ..... .



(a) 10 cm^2 (b) 20 cm^2 (c) 40 cm^2 (d) 80 cm^2

Scanned with OKEN Scanner

13 ✳ In the opposite figure, a rectangular coil of area A is placed in a magnetic field of flux density B such that the plane of the coil makes an angle of 60° with the field, where the magnitude of the magnetic flux passing through the coil equals 2×10^{-6} T.m^2, so the magnitude of the flux that is passing through the coil, when it rotates:



(i) In the clockwise direction:

① through 30°, equals ..... .

(a) 3.854×10^{-8} Wb
(b) 1.155×10^{-6} Wb
(c) 9.731×10^{-5} Wb
(d) 8.514×10^{-7} Wb

② through a quarter of a cycle, equals ..... .

(a) 6.25×10^{-6} Wb
(b) 7.93×10^{-4} Wb
(c) 1.155×10^{-6} Wb
(d) 9.11×10^{-4} Wb

(ii) In the counterclockwise direction:

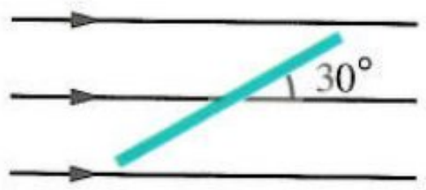
① through 30°, equals ..... .

(a) 1.155×10^{-6} Wb
(b) 2.31×10^{-6} Wb
(c) 4.692×10^{-7} Wb
(d) 3.854×10^{-8} Wb

② through a quarter of a cycle, equals ..... .

(a) 1.155×10^{-6} Wb
(b) 2.231×10^{-5} Wb
(c) 4.692×10^{-7} Wb
(d) 3.854×10^{-8} Wb

14 ✳ In the opposite figure, a coil of cross-sectional area A is placed in a magnetic field of flux density B, such that it makes an angle of 30° with the field, hence the total flux that is passing through the coil equals ϕ_m. The minimum angle by which the coil should rotate, so that the flux through it becomes:



(i) $2\,\phi_m$, equals ..... .

(a) 30°
(b) 45°
(c) 60°
(d) 90°

(ii) $\frac{2}{3}\,\phi_m$, equals ..... .

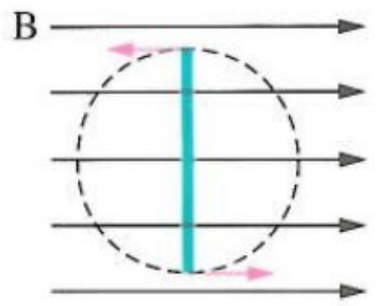
(a) 20.31°
(b) 10.53°
(c) 13.9°
(d) 15.52°

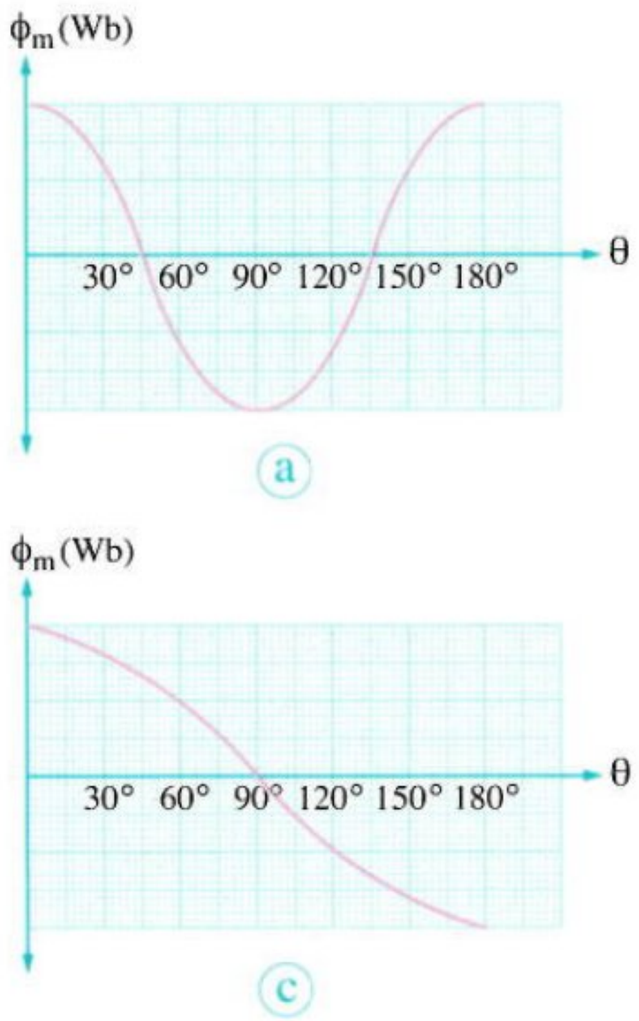
CamScanner

Scanned with OKEN Scanner

CamScanner

15 A rectangular coil rotates in a uniform magnetic field as in the opposite figure, hence the graph that represents the relation between the magnetic flux (ϕ_m) that penetrates the coil and the angle of rotation (θ) through which the coil rotates during a half cycle, starting from the perpendicular position to the magnetic field will be ..... .





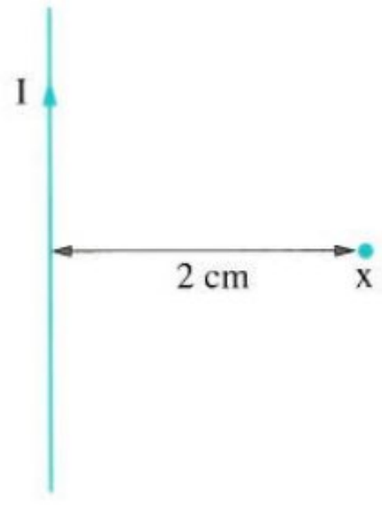
The magnetic field due to a current-carrying wire

16 The unit of $Wb.A^{-1}.m^{-1}$ is the measuring unit of ..... .

(a) magnetic flux
(b) magnetic flux density
(c) magnetic permeability
(d) electric power

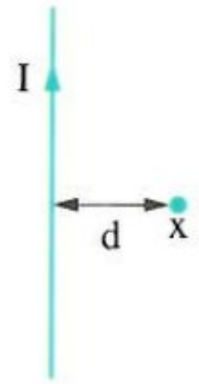
17 In the opposite figure, a long straight wire has a resistance of 0.2 Ω where the potential difference between its ends is 1 V, then the magnetic flux density at point x is ..... .

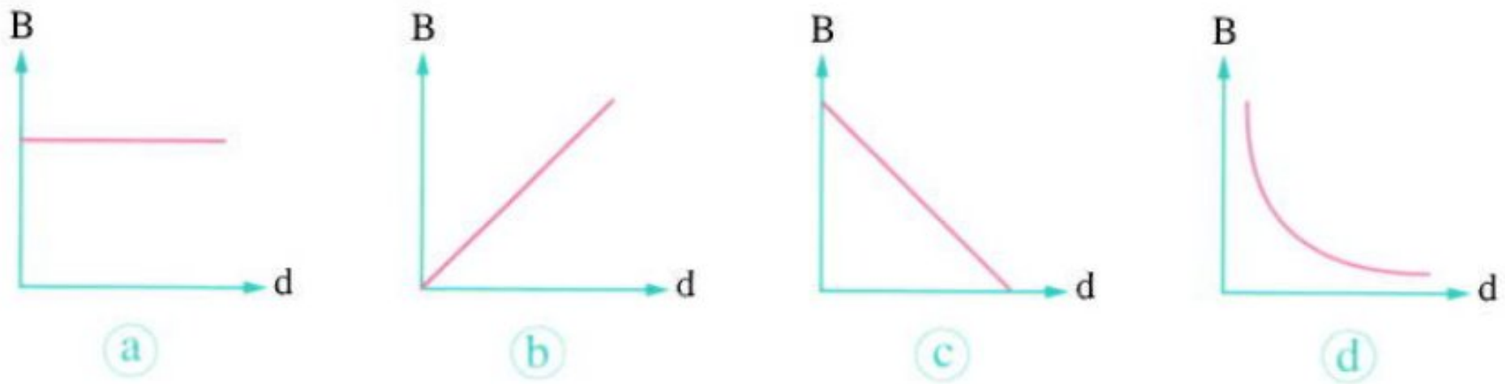
(a) 5×10^{-5} T and its direction is into the page
(b) 5×10^{-5} T and its direction is out of the page
(c) 2×10^{-5} T and its direction is into the page
(d) 2×10^{-5} T and its direction is out of the page



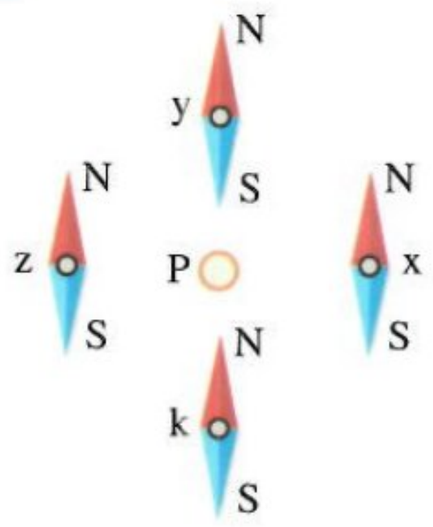
Scanned with OKEN Scanner

18 Which of the following graphs represents the relation between the magnetic flux density (B) at point x and the separation distance (d) of point x from the wire axis as shown in the opposite figure?



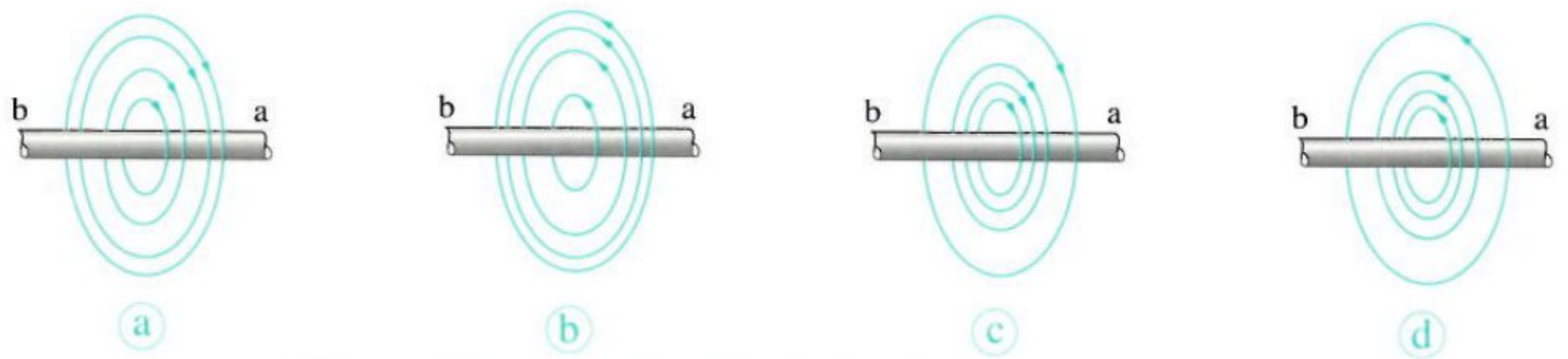


19 The opposite figure shows four compasses around a vertical straight wire (P), each needle of them is in the plane of the magnetic meridian, when the direction of the electric current passing through the wire is downward (into the page), a magnetic field is produced whose strength is greater than the magnetic field of the Earth, then the needle whose direction that refers to the north pole does not change is ..... .

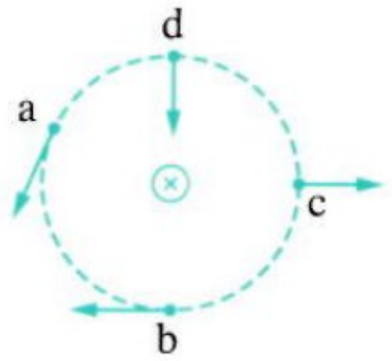


(a) x (b) y (c) z (d) k

20 A direct electric current passes through wire ab where the free electrons pass from the terminal a to terminal b, which of the following figures correctly represents the resulting magnetic flux lines around the wire?



21 The opposite figure shows a straight wire that is perpendicular to the plane of the page while carrying an electric current whose direction is into the page and the points a, b, c and d are located in the plane of the page and at equal distances from the wire, so the direction of the generated magnetic field due to the current of the wire is correct at point ..... .

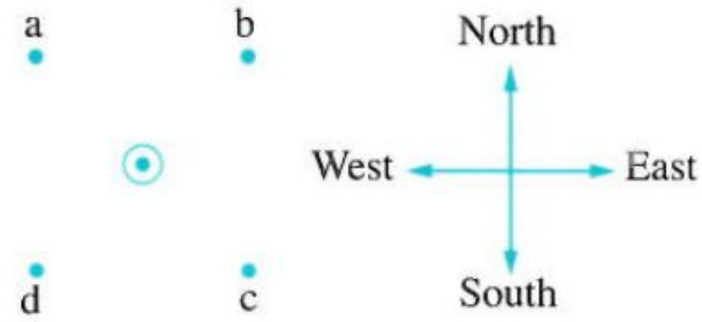


(a) a (b) b (c) c (d) d

CamScanner
Scanned with OKEN Scanner

CamScanner

22 The opposite figure shows a straight wire which is placed normal to the plane of the page while carrying an electric current that flows out of the page, hence the direction of the magnetic field will be towards the northwestern direction at point ..... .



(a) a (b) b (c) c (d) d

23 ✳ A straight wire is carrying a current of intensity 4 A. If the density of the magnetic flux at a certain point around the wire is 2×10^{-5} T, the separating distance of this point from the axis of the wire equals ..... .

(a) 0.01 m (b) 0.02 m (c) 0.03 m (d) 0.04 m

24 A straight wire carries an electric current of intensity I, if the produced magnetic flux at point whose perpendicular distance from the wire is 5 cm equals 5×10^{-5} T, the rate of electrons passing through a section of the wire is equal to ..... .

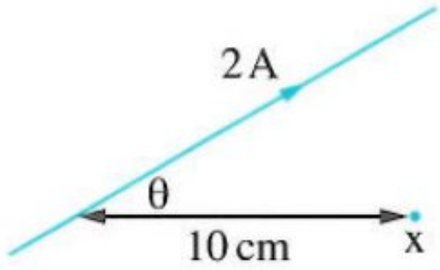
(a) 5.61×10^{21} electrons/s (b) 7.81×10^{21} electrons/s

(c) 5.61×10^{19} electrons/s (d) 7.81×10^{19} electrons/s

25 ✳ A battery of emf 8 V whose internal resistance is 2 Ω is connected to a straight wire of length 20 cm, cross-sectional area 3×10^{-8} m^2 and resistivity 4.5×10^{-6} Ω.m, so the magnetic flux density at a point that lies at a perpendicular distance of 10 cm away from the axis of the wire equals ..... .

(a) 4×10^{-6} T (b) 5×10^{-7} T (c) 6×10^{-8} T (d) 7×10^{-9} T

26 ✳ In the opposite figure, the magnetic flux density due to the flow of the electric current in the wire at point x is ..... .



(a) equal to 4×10^{-6} T (b) greater than 4×10^{-6} T

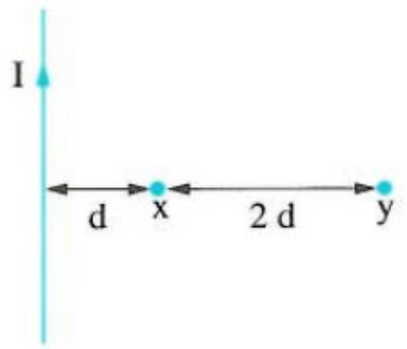
(c) smaller than 4×10^{-6} T (d) indeterminable

27 If the ratio between the magnetic flux densities at the points (X, Y) near a straight current-carrying wire is $\frac{B_X}{B_Y} = \frac{2}{3}$, so the ratio between the perpendicular distances of these two points from the wire $\frac{d_X}{d_Y}$ is ..... .

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{3}{2}$

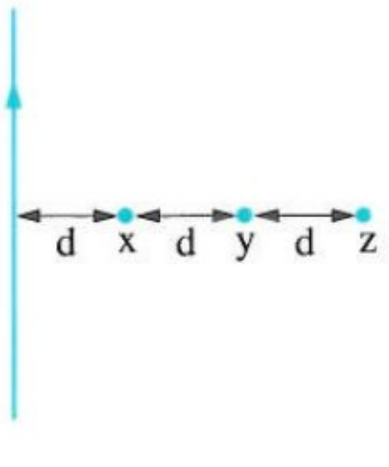
Scanned with OKEN Scanner

28 The opposite figure represents a straight wire that is carrying a direct electric current, so the ratio of the magnetic flux densities due to this current at point x to that at point y $\left(\frac{B_x}{B_y}\right)$ equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{3}{1}$ (c) $\frac{2}{1}$ (d) $\frac{1}{3}$

29 The opposite figure represents a long straight wire carrying a DC current, if the magnetic flux density at point z equals (B), then the magnetic flux density :



(i) at point (x) is ..... compared to the magnetic flux density at point (z).

(a) greater by $\frac{2}{3}$ B (b) less by $\frac{2}{3}$ B

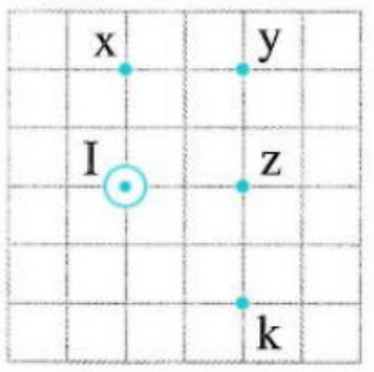
(c) greater by 2 B (d) less by 2 B

(ii) at point (y) is ..... compared to the magnetic flux density at point (z).

(a) greater by $\frac{1}{3}$ B (b) less by $\frac{1}{3}$ B

(c) greater by $\frac{1}{2}$ B (d) less by $\frac{1}{2}$ B

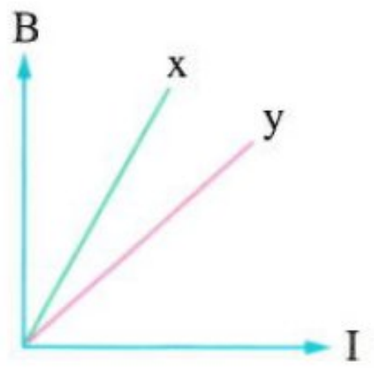
30 A straight wire is placed perpendicular to the page and carries a direct electric current that is directed out of the page as shown in the opposite figure, so the correct order for the value of the magneti c flux density at the shown points is .....



(a) $B_z > B_x = B_y > B_k$ (b) $B_y = B_z = B_k > B_x$

(c) $B_z > B_y = B_k > B_x$ (d) $B_z = B_x > B_y = B_k$

31 The opposite graph represents the relation between the magnetic flux density (B) which results due to the flow of a direct current in a straight wire and the intensity of this current (I) at two different points (x and y), so ..... .



(a) the distance of point x from the axis of the wire is larger than that of point y

(b) the distance of point x from the axis of the wire is less than that of point y

(c) the distance of point x from the axis of the wire is equal to that of point y

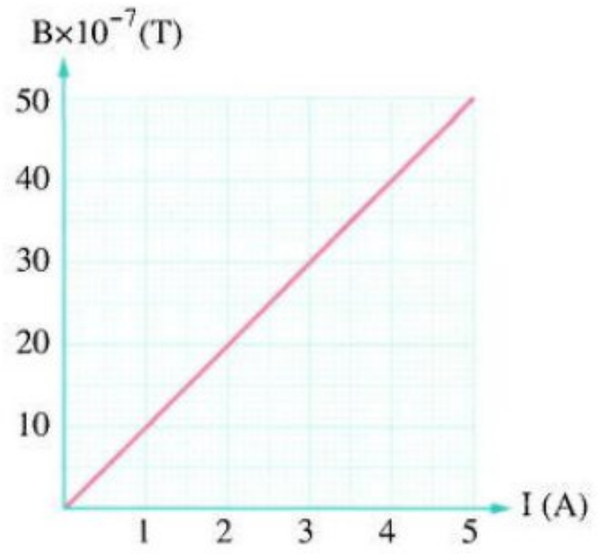
(d) the answer can't be determined

CamScanner

Scanned with OKEN Scanner

CamScanner

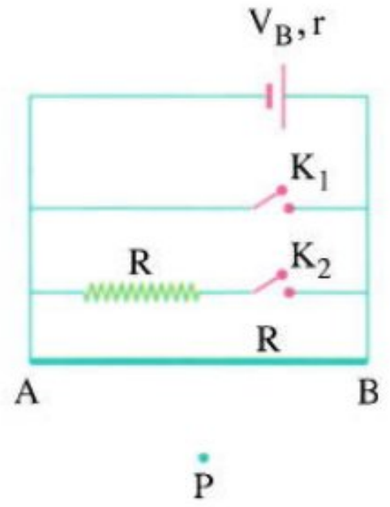
32 The opposite graph represents the relation between the magnetic flux density (B) at a definite point due to the flow of an electric current in a straight wire and the intensity of this current (I), so the separation distance between this point and the axis of the wire is ..... .



(a) 10 cm (b) 20 cm

(c) 40 cm (d) 100 cm

33 A straight wire AB of resistance R is connected to an electric circuit as shown in the opposite figure, so the magnetic flux density at point P due to the electric current passing through the wire AB has the maximum value when ..... .



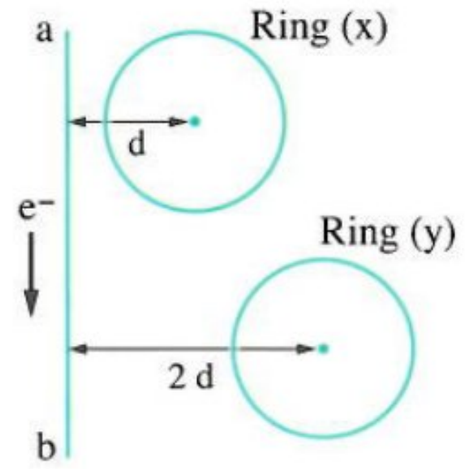
(a) the two switches K_1, K_2 are open

(b) the two switches K_1, K_2 are closed

(c) the switch K_1 is closed, the switch K_2 is open

(d) the switch K_1 is opened, the switch K_2 is closed

34 Two identical metal rings x, y are placed in the same vertical plane and next to them and in their plane is a straight wire ab as in the opposite figure, if the free electrons move through the wire from the terminal a to the terminal b, then :



(i) The direction of the magnetic flux lines that penetrate the two rings is ..... .

(a) perpendicular to the plane of the two rings and inward

(b) perpendicular to the plane of the two rings and outward

(c) in the plane of the two rings and towards the wire

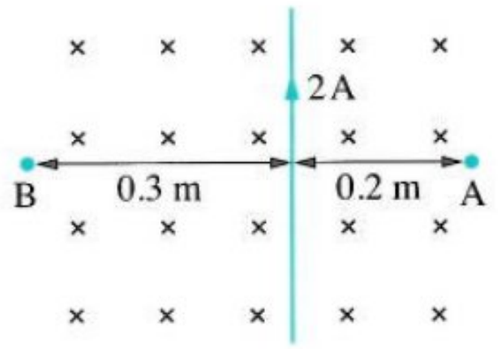
(d) in the plane of the two rings and away from the wire

(ii) The ratio between the values of the magnetic flux that passes through the two rings $\frac{(\phi_m)_x}{(\phi_m)_y}$ is ..... .

(a) greater than one (b) less than one (c) equal to one (d) can't be determined

Scanned with OKEN Scanner

35 ✳ In the opposite figure, a long straight wire carries an electric current of 2 A while being placed perpendicular to a uniform magnetic field of flux density 4×10^{-6} T, so the resultant magnetic flux density at:



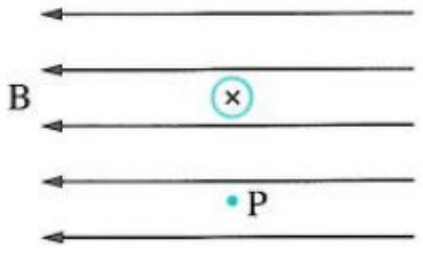
(i) point A equals ..... .

(a) 3.5×10^{-6} T (b) 6×10^{-6} T
(c) 8×10^{-6} T (d) 9×10^{-6} T

(ii) point B equals ..... .

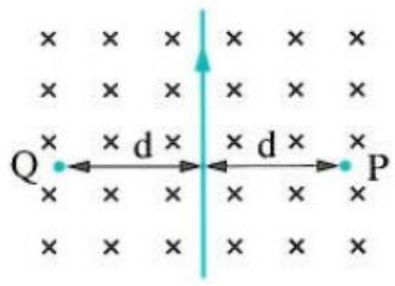
(a) 2.5×10^{-6} T (b) 5×10^{-6} T (c) 2.67×10^{-6} T (d) 9×10^{-6} T

36 ✳ In the opposite figure, a long straight wire carries an electric current of 40 A whose direction is perpendicular into the plane of the page. The wire is placed in a uniform magnetic field of 6×10^{-5} T that is directed to the left of the page, so the resultant magnetic flux density at point P which is 10 cm away from the axis of the wire is ..... .



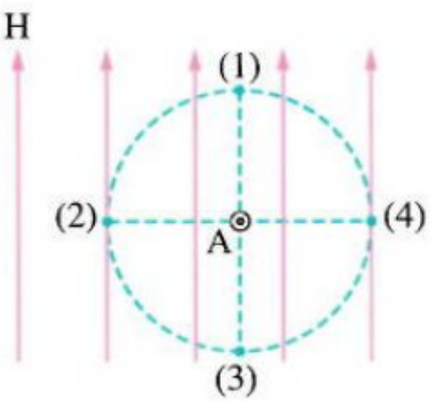
(a) 1.4×10^{-4} T (b) 1×10^{-4} T (c) 8×10^{-5} T (d) 2×10^{-5} T

37 ✳ In the opposite figure, a straight wire is placed perpendicular to a uniform magnetic field of flux density B that is directed into the page. When an electric current of intensity I passes in the wire, the total magnetic flux density at point P becomes 3 B, so the net magnetic flux density at point Q is ..... .



(a) zero (b) B (c) 2 B (d) 3 B

38 ✳ In the opposite figure, a straight wire (A) is placed perpendicular to the plane of the page while carrying an electric current that is directed out of the page, hence it produces a magnetic flux of density H tesla at the shown points on the circle. If the horizontal component of the magnetic flux density of Earth's magnetic field is H tesla in the direction shown, the resultant magnetic flux density at:



(i) point (1) equals ..... tesla.

(a) $\sqrt{2}$ H (b) $\sqrt{3}$ H (c) H (d) 2 H

(ii) point (2) equals ..... tesla.

(a) 0 (b) H (c) 3 H (d) $\sqrt{5}$ H

(iii) point (3) equals ..... tesla.

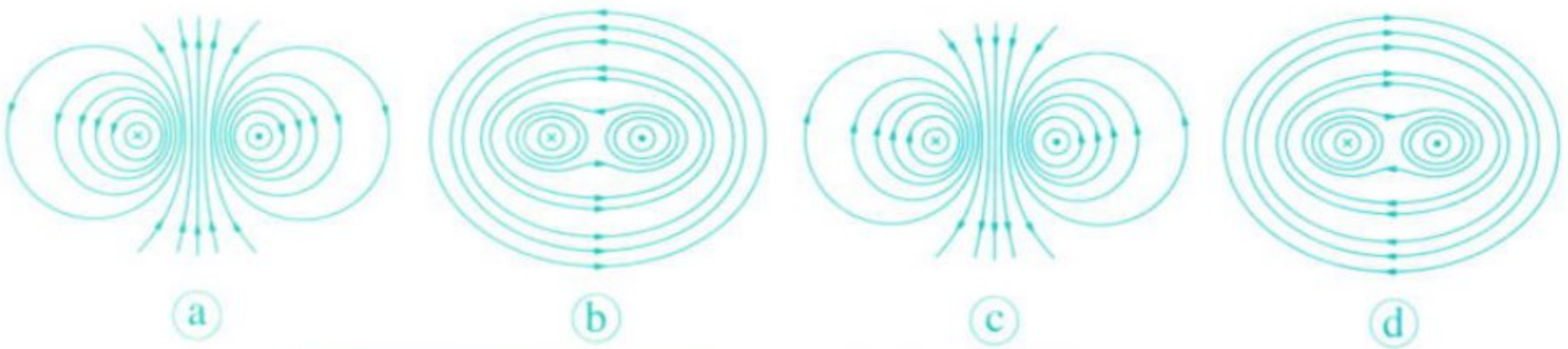
(a) 4 H (b) 2.5 H (c) $\sqrt{2}$ H (d) $\sqrt{7}$ H

(iv) point (4) equals ..... tesla.

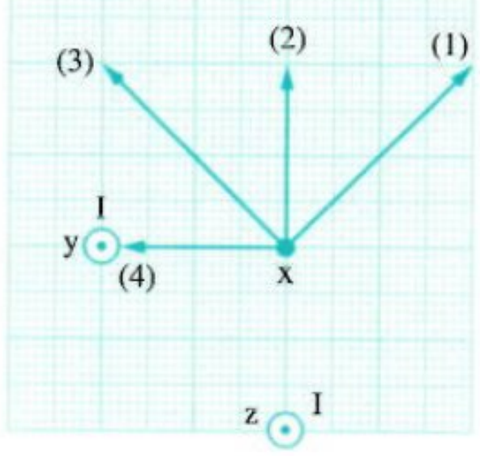
(a) H (b) 2 H (c) 3 H (d) 4 H

CamScanner
Scanned with OKEN Scanner

39 Which of the following diagrams correctly represents the magnetic flux lines pattern produced due to the flow of two equal electric currents in two opposite directions through two parallel wires?



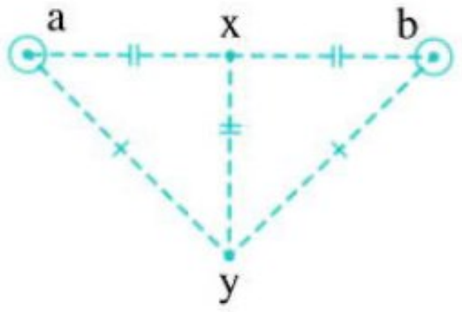
40 The opposite figure shows two parallel very long straight wires (y, z) that are carrying currents of equal intensities (I), hence the net magnetic flux density at point x is in the direction of ..... .



a (1) b (2)

c (3) d (4)

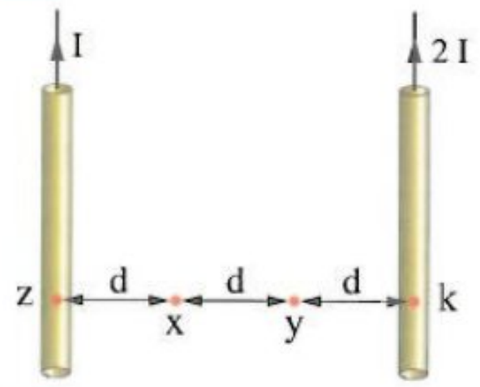
41 In the opposite figure, two parallel straight wires a, b which are perpendicular to the plane of the page carry electric currents I and 2 I respectively, so the net magnetic flux density (B) at point y is calculated from the relation; .....



a $B = B_a + B_b$ b $B = B_a - B_b$

c $B = B_b - 2\,B_a$ d $B = \sqrt{B_a^2 + B_b^2}$

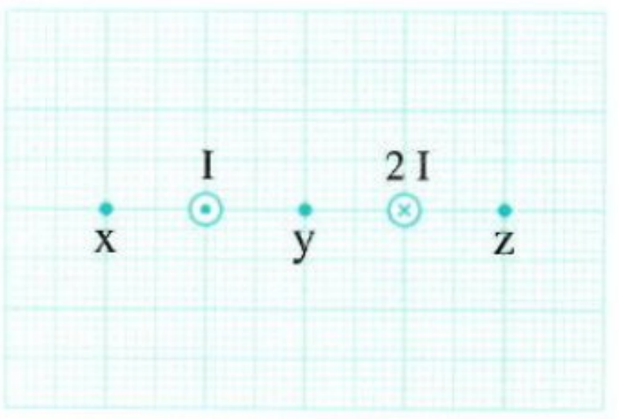
42 The opposite figure shows two long parallel straight wires where each of them carries an electric current, so the magnetic flux density is greater at point ..... .



a x b y

c z d k

43 Two anti-parallel electric currents of intensities I, 2 I pass through two long straight wires as shown in the opposite figure, so the correct order of magnetic flux densities at points x, y and z is .....

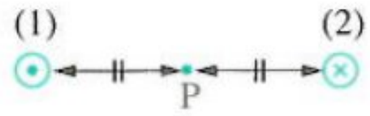


a $B_x > B_y > B_z$ b $B_z > B_y > B_x$

c $B_y > B_x > B_z$ d $B_y > B_z > B_x$

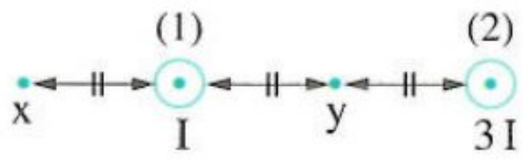
Scanned with OKEN Scanner

44 The opposite figure shows two very long parallel straight wires that are perpendicular to the page while each of them carries an electric current. If the electric current through the first wire is greater than that in the second wire, then the direction of the resultant magnetic flux at point P is in the plane of the page and to the ..... .



(a) right (b) left (c) upward (d) downward

45 In the opposite figure, two long straight parallel wires (1) , (2) carry electric currents of intensities I and 3 I respectively. When comparing the resultant magnetic flux densities at points x and y, we find that the resultants ..... .



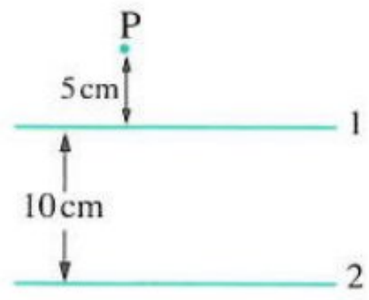
(a) have the same magnitude and direction

(b) are different in both magnitude and direction

(c) have the same magnitude and differ in direction

(d) differ in magnitude but have the same direction

46 * The opposite figure shows two parallel wires, where wire 1 carries a current of intensity 2 A and wire 2 carries a current of 4 A, so the total magnetic flux density at:



(i) point P, if the two currents are in the same direction, equals ..... .

(a) 1.33×10^{-5} T (b) 2.66×10^{-5} T (c) 4.43×10^{-6} T (d) 6.65×10^{-6} T

(ii) point P, if the two currents are in opposite directions, equals ..... .

(a) 4.14×10^{-5} T (b) 3.06×10^{-7} T (c) 2.9×10^{-8} T (d) 2.67×10^{-6} T

47 * If 2.5×10^{20} electrons pass every one second through a cross-section of a straight wire x which is placed parallel to another straight wire y that is at a distance of 5 cm and carries a current of 40 A, so the magnetic flux density at the mid-point between the two wires when the electrons motion in wire x and the direction of the conventional current in wire y:

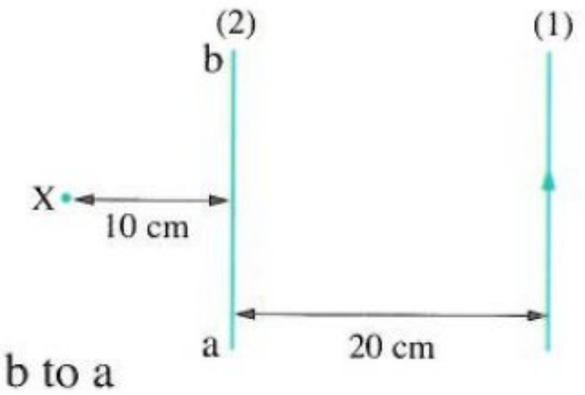
(i) are in the same direction, equals ..... .

(a) 1.6×10^{-4} T (b) 0 (c) 2.56×10^{-4} T (d) 6.4×10^{-4} T

(ii) are in opposite directions, equals ..... .

(a) 0 (b) 6.4×10^{-4} T (c) 3.2×10^{-4} T (d) 1.92×10^{-4} T

48 The opposite figure shows two parallel long wires (1), (2) which are separated by a distance of 20 cm in the same plane of the page. Wire (1) carries a current of intensity 9 A, so the magnitude and the direction of the current in wire (2) that makes the net magnetic field at point X vanish are ..... .



(a) 4.5 A, from a to b (b) 4.5 A, from b to a

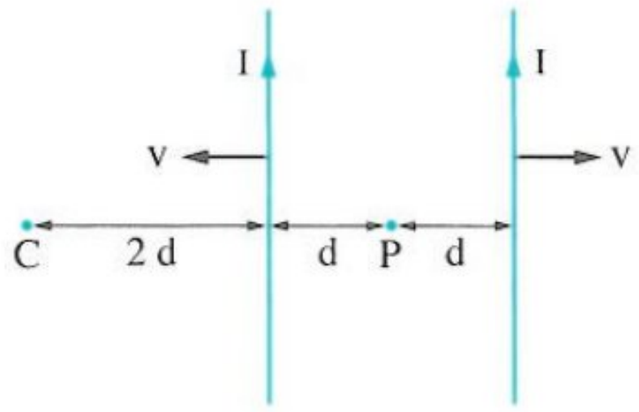
(c) 3 A, from a to b (d) 3 A, from b to a

CamScanner

Scanned with OKEN Scanner

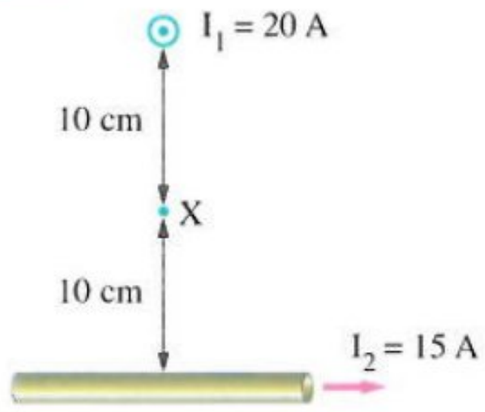
CamScanner

49 Two parallel straight wires carry an electric current of intensity I in the same direction while the two wires are moved a distance d in two opposite directions at the same instant with a uniform velocity (v) as shown in the opposite figure, so the resultant magnetic flux at each of the two points P, C respectively ..... .



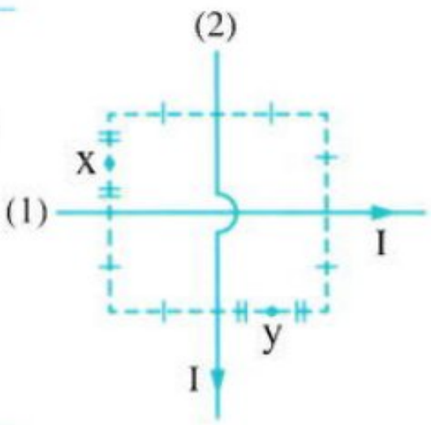
(a) remains zero, increases
(b) remains zero, decreases
(c) increases, increases
(d) increases, decreases

50 In the opposite figure, two long straight wires are perpendicular to each other, while the shortest distance between them is 20 cm. If the first wire carries 20 A perpendicularly out of the page and the second wire carries a current of 15 A in a direction parallel to the page, the net magnetic flux density at point X will be ..... .



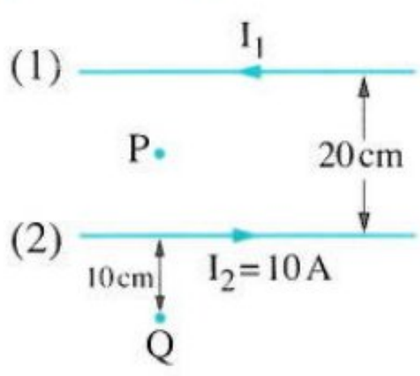
(a) 3×10^{-5} T
(b) 4×10^{-5} T
(c) 5×10^{-5} T
(d) 6×10^{-5} T

51 * The opposite figure shows two insulated perpendicular straight wires, each carrying a current of intensity I, so the ratio between the two flux densities at the two points x and y respectively is ..... .



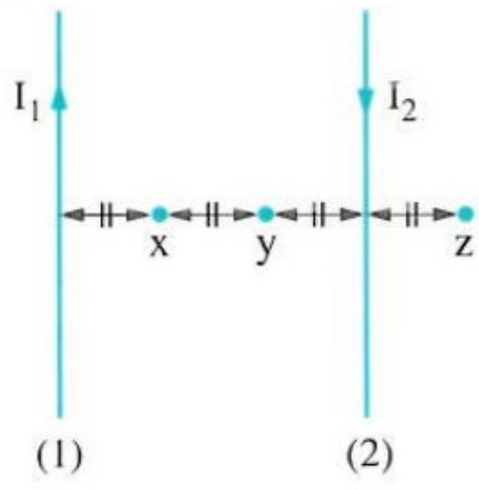
(a) 1 : 1
(b) 1 : 2
(c) 2 : 1
(d) 2 : 3

52 * The opposite figure shows two parallel straight wires (1), (2). If the total magnetic flux density B_t at point P (in the middle of the distance between the two wires) equals 6×10^{-5} T, the total magnetic flux density at point Q equals ..... .



(a) 3.35×10^{-5} T
(b) 2.68×10^{-6} T
(c) 2.01×10^{-5} T
(d) 6.7×10^{-6} T

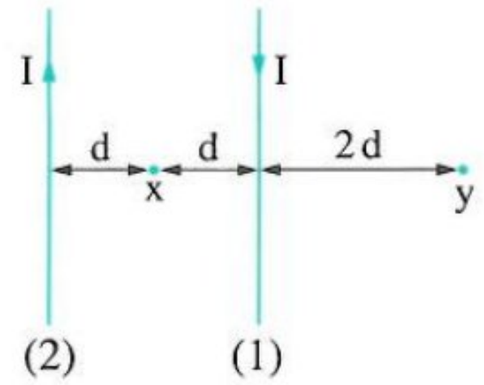
53 * In the opposite figure, two long straight parallel wires (1) , (2) carry electric currents. If the resultant magnetic flux densities at the two points x and y are equal, and the resultant magnetic flux density at point z is (B), then the resultant magnetic flux density at point x is ..... .



(a) 0.75 B
(b) B
(c) 1.5 B
(d) 2 B

Scanned with OKEN Scanner

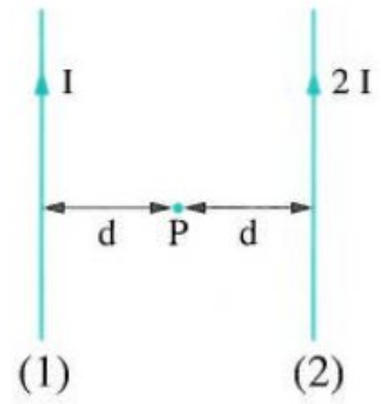
54 ✳ If the magnetic flux density at point x is B, then the magnetic flux density at point y is ..... .



(a) $\frac{B}{12}$ (b) 12 B

(c) $\frac{B}{2}$ (d) $\frac{B}{8}$

55 ✳ The opposite figure represents two very long parallel straight wires where each of them carries an electric current. If the magnetic flux density due to the current of wire (1) at point P equals B, so the resultant and the direction of the magnetic flux density at point P are ..... .



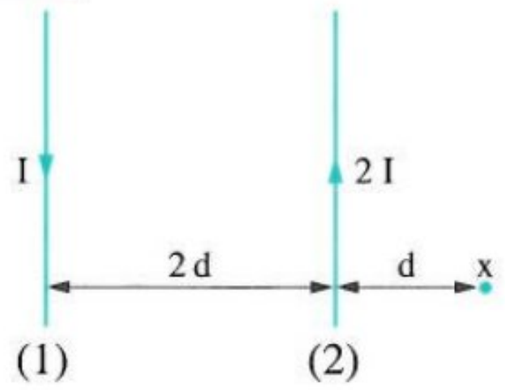
(a) B, perpendicular into the page

(b) 3 B, perpendicular into the page

(c) B, perpendicular out of the page

(d) 3 B, perpendicular out of the page

56 In the opposite figure, when the electric current intensity in wire (1) increases to 6 I, the resultant magnetic flux density at point x ..... .



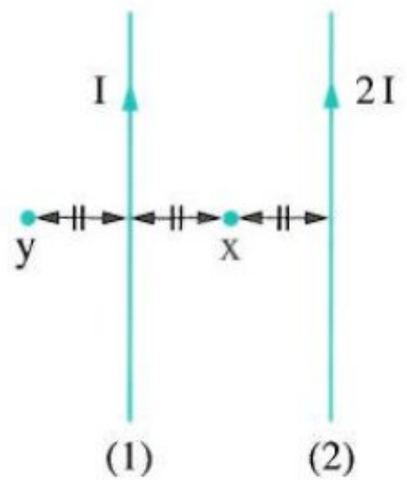
(a) increases to three times its initial value

(b) increases to six times its initial value

(c) becomes zero

(d) decreases to one third its initial value

57 Two long straight parallel wires (1) and (2) carry electric currents in the same direction with intensities I and 2I respectively as shown in the opposite figure, which of the following changes leads to an increase in the magnetic flux density at both points x and y?



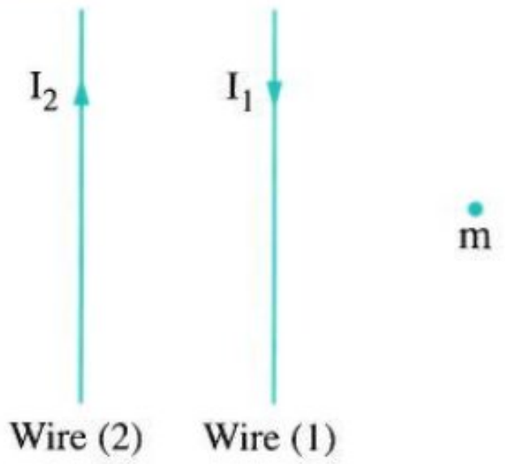
(a) The current intensity increases in wire 1 to 2 I

(b) The current intensity increases in wire 1 to 3 I

(c) The current intensity increases in wire 2 to 4 I

(d) No current passes through wire 2

58 In the opposite figure, two long straight parallel wires (1) and (2), carry electric currents in two opposite directions where $I_1 > I_2$. Which of the following changes leads to an increase in the magnitude of the resultant magnetic flux density at point m?



(a) The current intensity I_2 increases until it equals I_1

(b) The current intensity I_1 decreases until it equals I_2

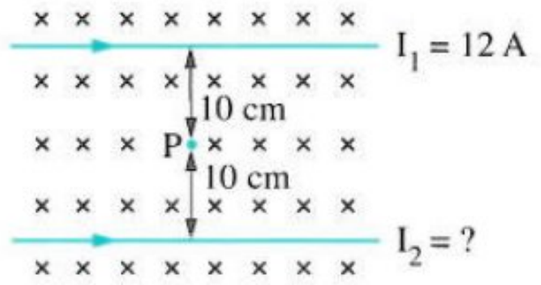
(c) The direction of the current in one of the wires is reversed.

(d) Bringing one wire closer to the other.

CamScanner

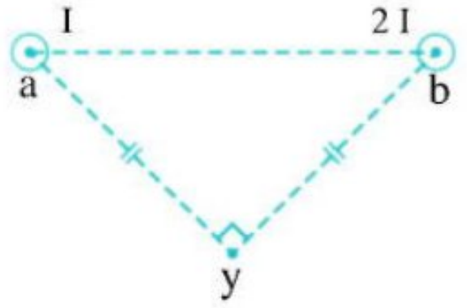
Scanned with OKEN Scanner

59 The opposite figure shows two very long and parallel straight wires that are placed in the plane of the page affected by a uniform external magnetic field of flux density 2×10^{-5} T whose direction is perpendicular into the plane of the page, if the resultant magnetic flux density at point P equals 2×10^{-5} T and its direction is into the page, so the electric current intensity that passes through the second wire equals ..... .



(a) 24 A (b) 18 A (c) 12 A (d) 6 A

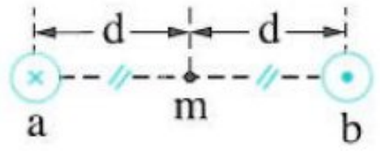
60 In the opposite figure, two straight parallel wires a and b carry electric currents I and 2 I respectively, as shown. If the formed flux density due to wire a at point y is B, the net flux density at point y equals ..... .



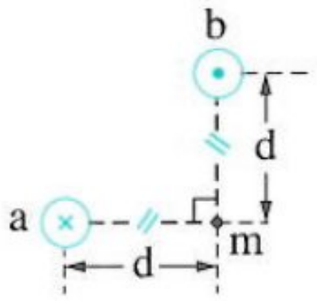
(a) B (b) 2 B

(c) $\sqrt{3}$ B (d) $\sqrt{5}$ B

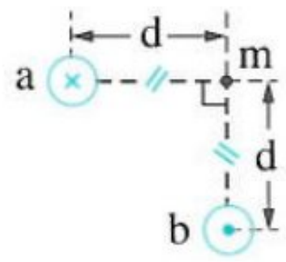
61 Two long straight parallel wires a and b are placed perpendicular to the plane of the page and carry two electric currents of the same intensity and in two opposite directions, if the wires are placed in different positions relative to each other as shown in the following figures:



(1)



(2)



(3)

Then, at point (m),

(i) the resultant magnetic flux density is ..... .

(a) greater in figure (1) (b) greater in figure (2)

(c) greater in figure (3) (d) equal in the three figures

(ii) the resultant magnetic flux densities in figures (2) and (3) ..... .

(a) have the same magnitude and direction

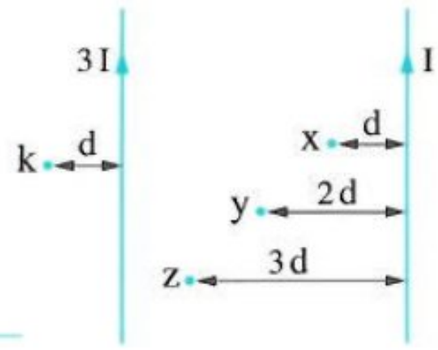
(b) have the same magnitude but differ in direction

(c) differ in magnitude but have the same direction

(d) differ in magnitude and in direction

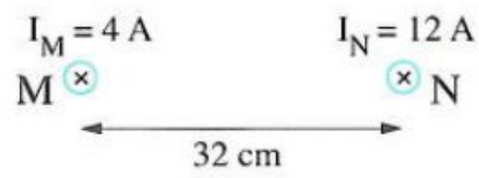
Scanned with OKEN Scanner

62 In the opposite figure, if the distance between the two wires equals 4 d. So, the neutral point is point ..... .



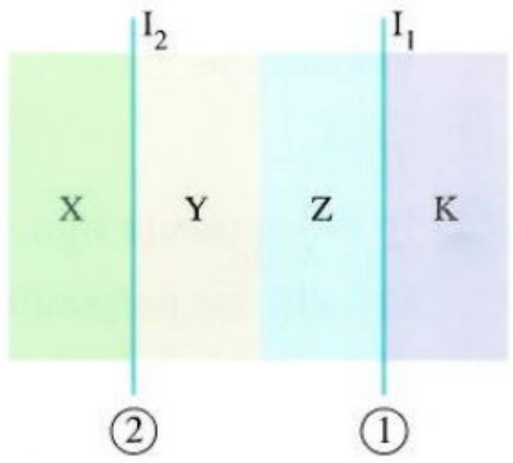
(a) x (b) y
(c) z (d) k

63 The opposite figure shows two long parallel wires M and N. If the two wires are perpendicular to the page and carry currents that flow into the page, the neutral point will be ..... .



(a) between them at a distance 8 cm from wire M
(b) outside of them at a distance 8 cm from wire M
(c) between them at a distance 24 cm from wire M
(d) outside of them at a distance 24 cm from wire M

64 * The opposite figure represents two long straight parallel wires in the same plane, each of them carrying an electric current where $I_1 > I_2$. So, in which of the regions X, Y, Z, K, the neutral point lies if the currents which are passing in the wires are:



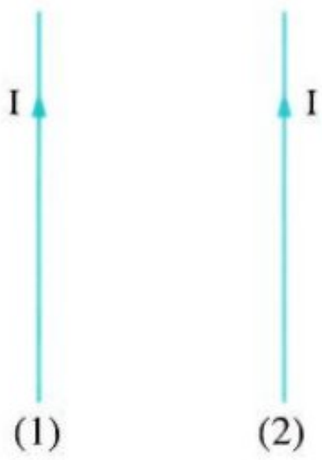
(i) In the same direction?

(a) X (b) Y (c) Z (d) K

(ii) In opposite directions?

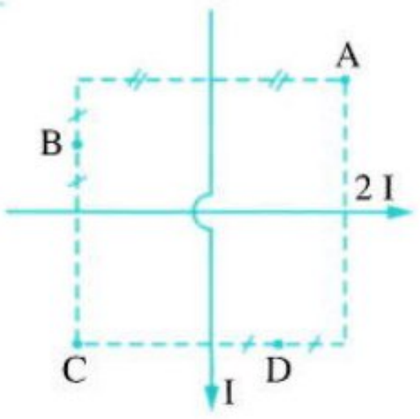
(a) X (b) Y (c) Z (d) K

65 Two parallel straight wires (1), (2) carry an electric current of intensity I in the same direction, which of the following procedures leads to the disappearance of the neutral point?



(a) One of the two wires is displaced away from the other.
(b) The electric current intensity is doubled in one of the two wires.
(c) The direction of the current is reversed in one of the two wires.
(d) The direction of the current is reversed in each of the two wires.

66 In the opposite figure, two insulated perpendicular wires carry electric currents, so the net magnetic flux density vanishes at point ..... .

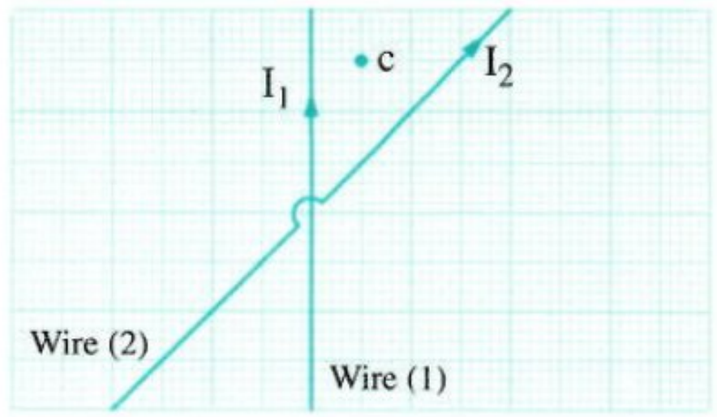


(a) A (b) B
(c) C (d) D

CamScanner
Scanned with OKEN Scanner

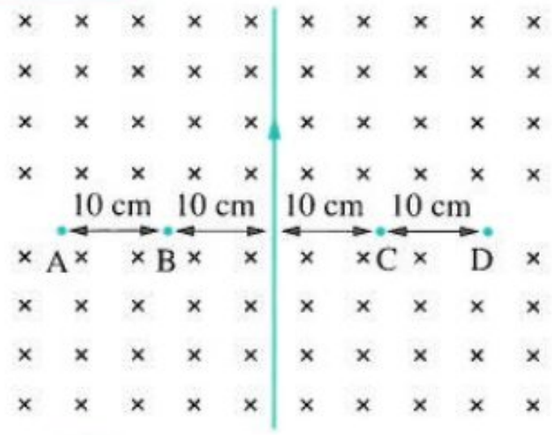
CamScanner

67 Two long insulated straight wires 1 and 2, carry electric currents with intensities I_1 and I_2, and their directions are as shown in the opposite figure.
If point c represents the neutral point, which of the following relations is correct regarding the current intensities of the two wires?



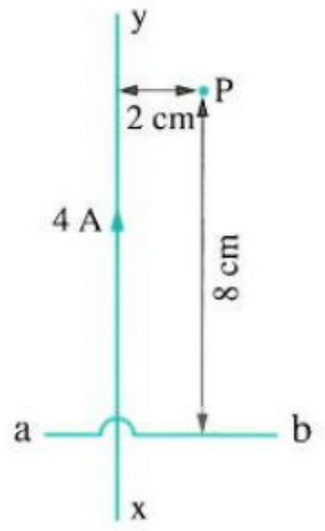
(a) $I_1 > I_2$ (b) $I_1 < I_2$
(c) $I_1 = I_2$ (d) Indeterminable.

68 In the opposite figure, a long straight wire that carries an electric current of intensity 10 A, is placed in the plane of the page perpendicular to a uniform magnetic field of flux density 2×10^{-5} T that is directed into the page, then the point at which the resultant magnetic flux density vanishes is ..... .



(a) A (b) B (c) C (d) D

69 In the opposite figure, two very long straight and isolated wires (xy, ab) are perpendicular to each other. Wire (xy) carries a current of intensity 4 A, so the intensity and the direction of the current that has to be set up in wire (ab) so that the resultant magnetic flux density at point P vanishes are ..... .

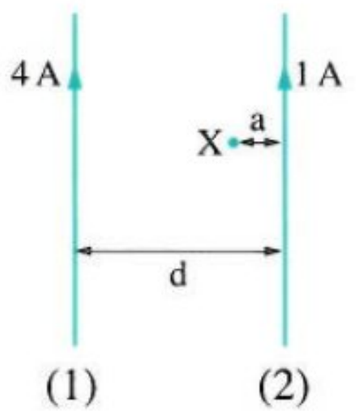


(a) 8 A, from a to b (b) 8 A, from b to a
(c) 16 A, from a to b (d) 16 A, from b to a

70 ✱ A small compass is placed at a point between two straight parallel wires where it lies at a distance of 20 cm from the first wire which carries a current of intensity 2 A, directed from south to north and at a distance of 40 cm from the second wire, so the intensity and the direction of the current in the second wire that don't make the compass needle deflect are ..... .

(a) 4 A, from north to south (b) 2 A, from north to south
(c) 4 A, from south to north (d) 2 A, from south to north

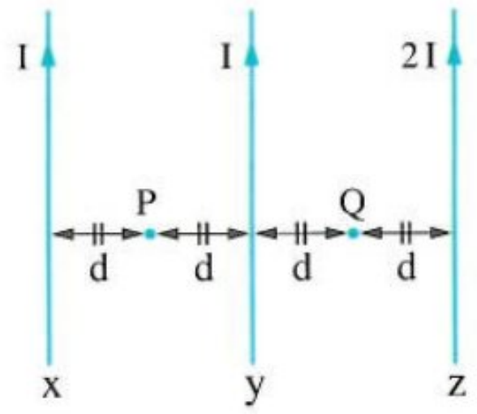
71 ✱ The opposite figure shows two parallel straight wires (1) and (2) that carry currents where point X lies at the neutral point and away from wire (2) by a distance a, hence if the intensity of the electric current in wire (2) is increased to 4 A, the neutral point has moved a distance of 10 cm, so the distance d between the axes of the two wires equals ..... .



(a) 18.51 cm (b) 20.83 cm (c) 24.75 cm (d) 33.33 cm

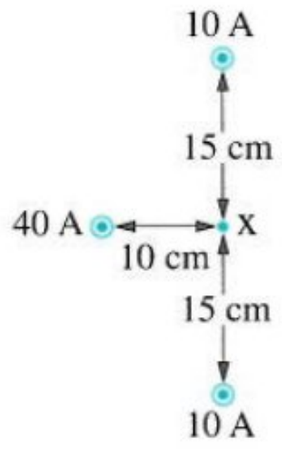
Scanned with OKEN Scanner

72 In the opposite figure, three long straight wires are parallel and each of them carries an electric current, so the directions of the magnetic fields at points P and Q are ..... .



	P	Q
a	out of the page	out of the page
b	into the page	into the page
c	out of the page	into the page
d	into the page	out of the page

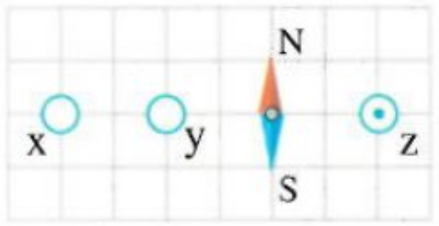
73 The opposite figure represents three wires that are perpendicular to the plane of the page, each of them carries an electric current as in the figure, then the net magnetic flux density at point x equals ..... .



(a) 2×10^{-5} T (b) 8×10^{-5} T

(c) 9×10^{-5} T (d) 4×10^{-4} T

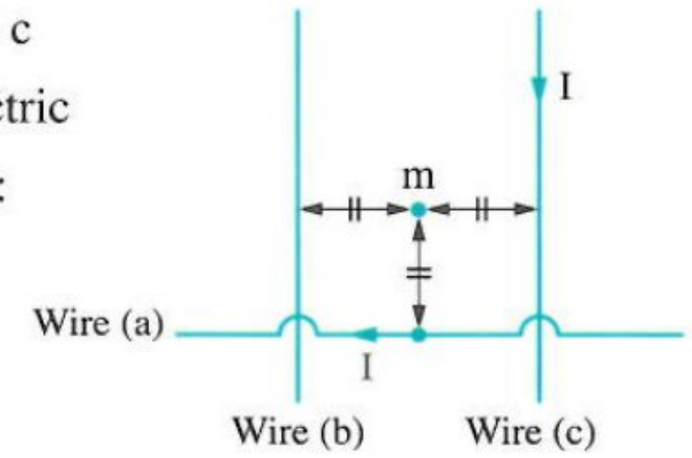
74 Three vertical straight wires x, y and z are in one plane and carry an electric current of intensity I, a compass is placed between two wires y, z, then its needle takes the position shown in the opposite figure, by ignoring the effect of the magnetic field of the Earth, then the directions of the currents in the two wires x, y respectively are ..... .



(a) out of the page, out of the page (b) out of the page, into the page

(c) into the page, out of the page (d) into the page, into the page

75 In the opposite figure, three long straight wires a, b and c are placed in one plane and each of them carries an electric current, if the point m represents the neutral point, then:

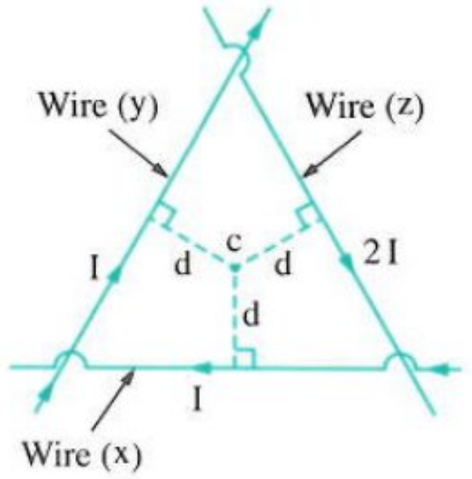


	The direction of the current in wire (b)	The intensity of the current in wire (b)
a	Upward	I
b	Upward	2I
c	Downward	I
d	Downward	2I

CamScanner
Scanned with OKEN Scanner

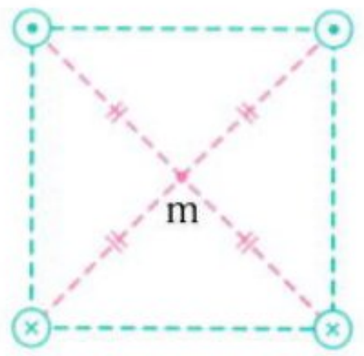
CamScanner

76 ✱ In the opposite figure, three current-carrying wires x, y, z are in the same plane. If the magnetic flux density due to the current in wire z at point c equals B_o, then the value of the net magnetic flux density at point c which is due to the currents in the all three wires is equal to ..... .



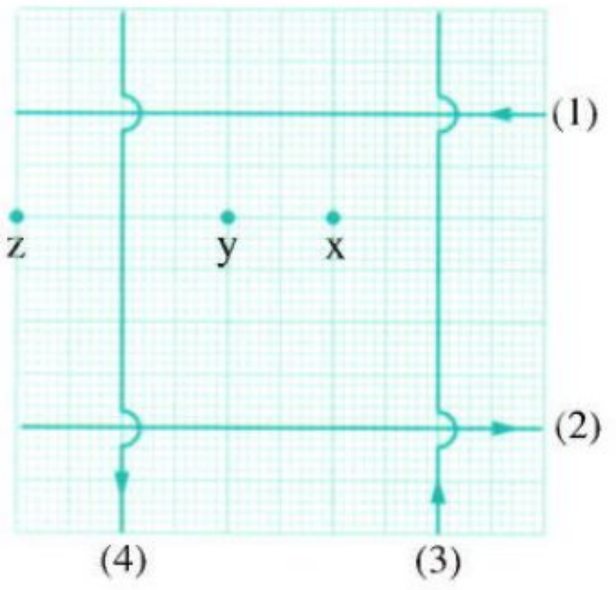
(a) zero
(b) $2\ B_o$
(c) $3\ B_o$
(d) $4\ B_o$

77 In the opposite figure, four very long straight parallel wires are placed perpendicular to the plane of the page at the vertices of a square, if all of them carry electric currents of the same intensity that flow in directions as shown in the figure, then the direction of the resultant magnetic flux density at point (m) is ..... .



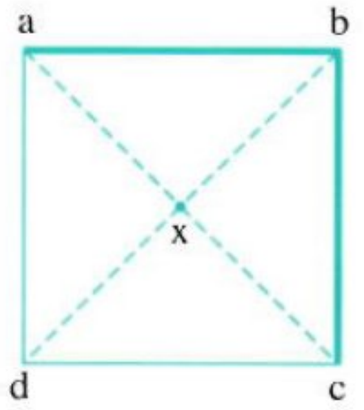
(a) perpendicular into the page
(b) perpendicular out of the page
(c) in the plane of the page to the right
(d) in the plane of the page to the left

78 The opposite diagram shows four long straight wires (1, 2, 3 and 4) in the same plane, each carrying an electric current of intensity I. Then the correct order of the points x, y and z in terms of the resultant magnetic flux density at each of them is ..... .



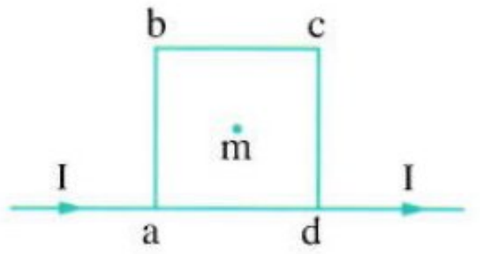
(a) $z < y < x$
(b) $z > y > x$
(c) $z < y = x$
(d) $z > y = x$

79 Two wires of the same metal abc, adc have the same length and different in the cross-sectional area, they are connected together to form a square as shown in the opposite figure, so the two points which when connected to an electric source, make the resultant magnetic flux density at point x equal to zero are ..... .



(a) a, d
(b) a, c
(c) a, b
(d) b, d

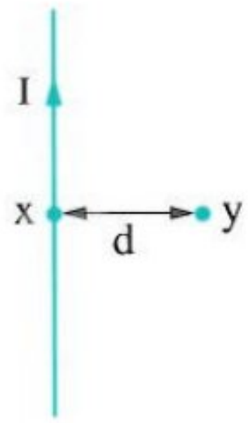
80 ✱ A wire of uniform cross-section is shaped in the form of a square abcd and an electric current passes through it as shown in the opposite figure. If the magnetic flux density at point (m) caused by the current passing through only segment (ab) of the wire equals (B_o), then the resultant magnetic flux density at point (m) equals ..... .



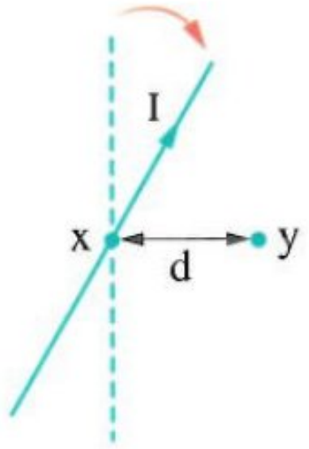
(a) zero
(b) B_o
(c) $2B_o$
(d) $4\ B_o$

Scanned with OKEN Scanner

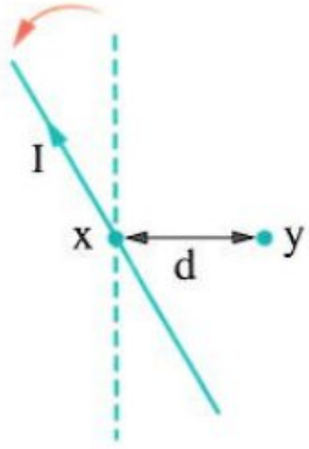
81 ✱ A long straight wire lies in the plane of the page, carrying an electric current of intensity I and away from which a point y is located at distance d as in figure (1). If the wire is rotated in the plane of the page about an axis passing through point x by 30°, once in the clockwise direction as shown in figure (2), and another time in the counter-clockwise direction as shown in figure (3).



(1)



(2)

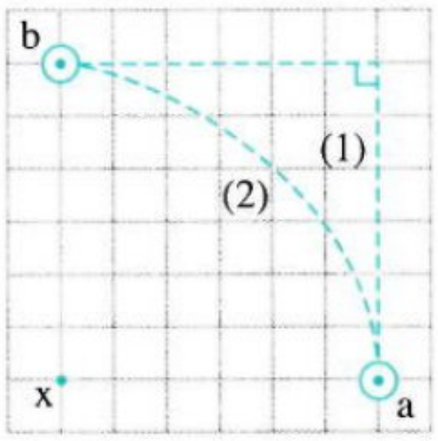


(3)

Then, the correct order for the magnitudes of the magnetic flux density at point (y) in the three figures is .....

(a) $B_1 = B_2 = B_3$
(b) $B_1 > B_2 = B_3$
(c) $B_2 > B_3 > B_1$
(d) $B_2 = B_3 > B_1$

82 A straight wire carries a direct electric current, is placed perpendicular to the page and it can be moved from position a to position b through two paths (1) or (2) (where path (2) is a circular path, its center is point x) as shown in the opposite figure, what happens to the value of the magnetic flux density at point x during the transmission of the wire through:



(i) path (1)?

(a) Increases.
(b) Doesn't change.
(c) Increases then decreases.
(d) Decreases then increases.

(ii) path (2)?

(a) Increases.
(b) Doesn't change.
(c) Increases then decreases.
(d) Decreases then increases.

Second Essay questions

Magnetic flux and flux density

1 A straight wire of length 1 m is coiled to form a loop of one turn and placed perpendicular to a magnetic field of flux density 1 T. **Find** the magnitude of the magnetic flux that passes through the loop if it is:

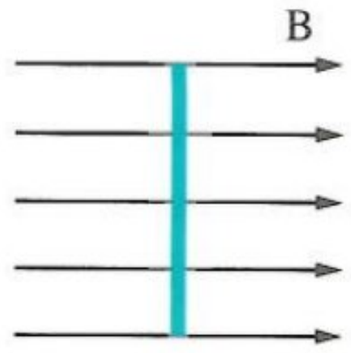
(a) Circular.
(b) Square.

CamScanner
Scanned with OKEN Scanner

CamScanner

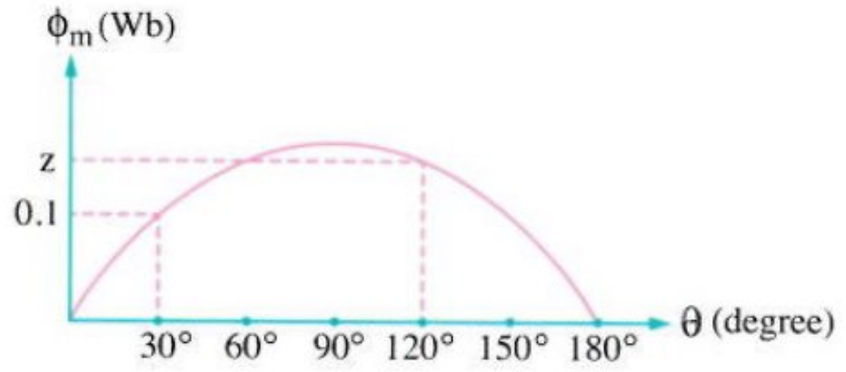
2 The opposite figure shows a lateral view of a coil that was placed in a uniform magnetic field, so if the value of the flux that is penetrating the coil is 4×10^{-2} Wb, **what** will be the least angle by which the coil rotates in the clockwise direction from the shown position in the figure so that the flux penetrating the coil becomes:

(a) 2×10^{-2} Wb? (b) -2×10^{-2} Wb?



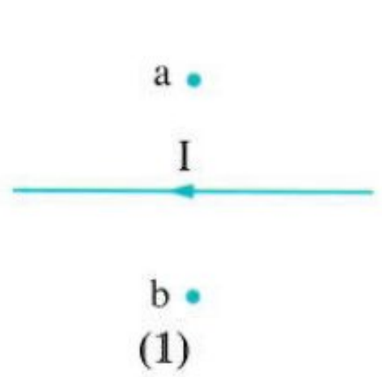
3 The opposite graph represents the relation between the magnetic flux (ϕ_m) that penetrates a coil while rotating with a uniform rate inside a magnetic field and the angle (θ) between the plane of the coil and the field.

Calculate the value of z.

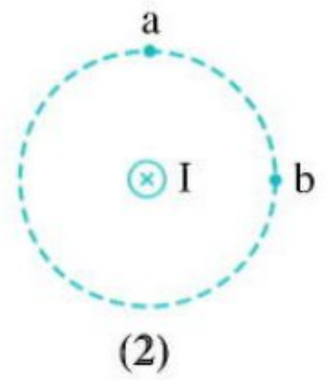


The magnetic field due to a current-carrying wire

4 **What** is the direction of the magnetic field due to the electric current (I) in each of the straight wires shown in figures (1), (2) at each of the points a, b?

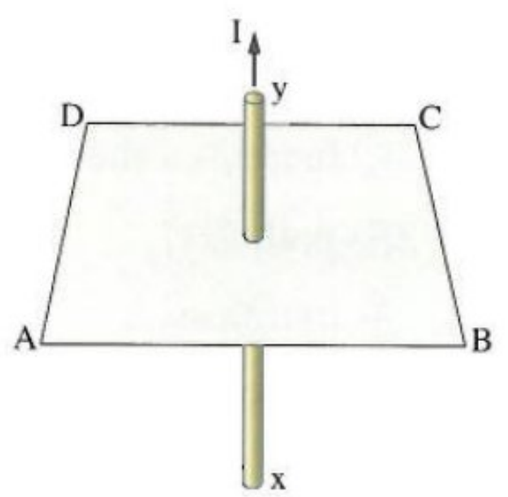


(1)



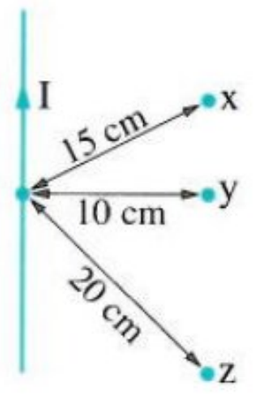
(2)

5 In the opposite figure, a vertical straight wire xy that carries an electric current penetrates a horizontal cardboard sheet. **Draw on the sheet** ABCD four of the lines of the magnetic field that arises due to the current in wire xy.



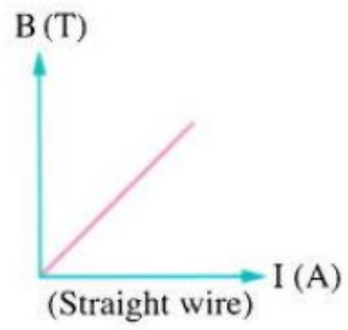
6 In the opposite figure, a very long straight wire carries an electric current where three points x, y, z lie on a parallel line beside it.

Compare the values of the magnetic flux densities at these points, **explaining your answer**.

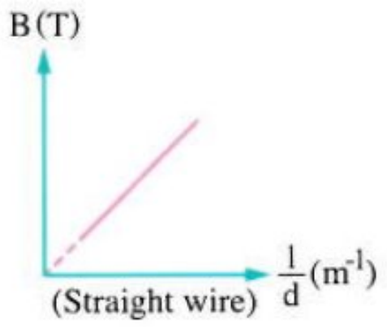


Scanned with OKEN Scanner

7 **Write the mathematical relations and the slope of the lines that are represented in the following graphs:**



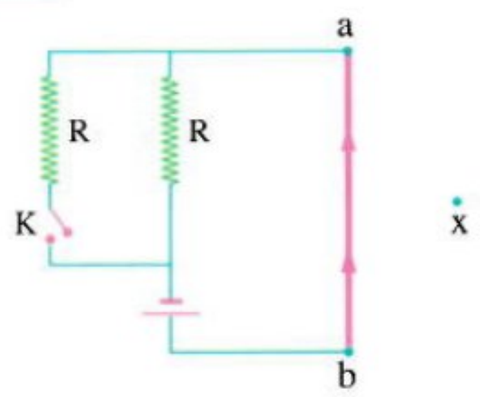
(a)



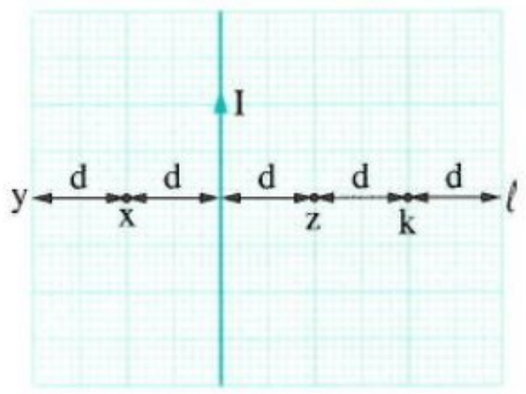
(b)

Where: (B) is the magnetic flux density, (I) is the electric current intensity in the wire, (d) is the distance between the point and the wire axis.

8 **How can you explain:** The value of the magnetic flux density at point x due to the current in wire ab increases when switch K gets closed in the opposite circuit.



9 In the opposite figure, if the magnetic flux density due to the flow of electric current (I) in the wire at point x equals B in a certain direction, **determine with explanation** the point at which the magnetic flux density becomes $\frac{B}{2}$ in the opposite direction.



10 Figure (1) shows a compass that is placed in the plane of magnetic meridian below a straight wire. When an electric current passed in the wire, the compass needle was deflected to settle as in the figure (2). **Determine** the direction of the electric current passing in the wire.

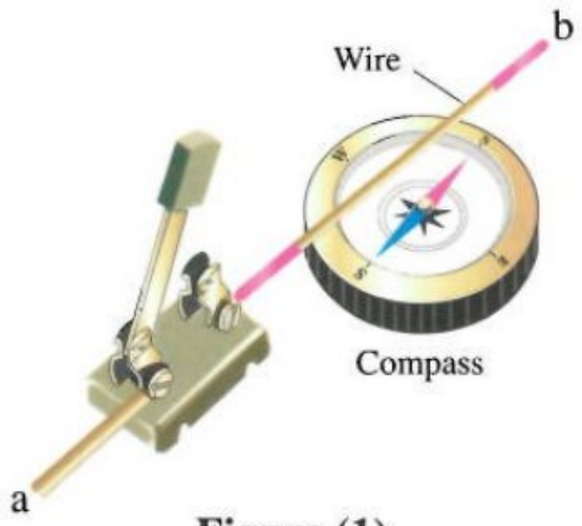


Figure (1)

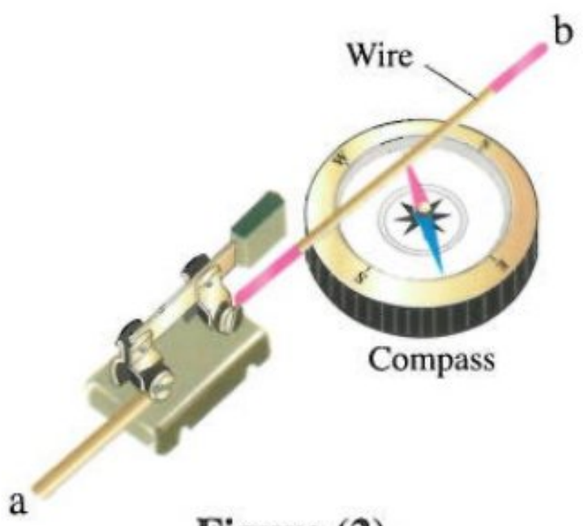


Figure (2)

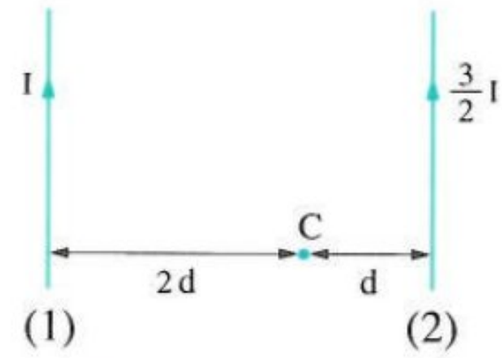
CamScanner
Scanned with OKEN Scanner

CamScanner

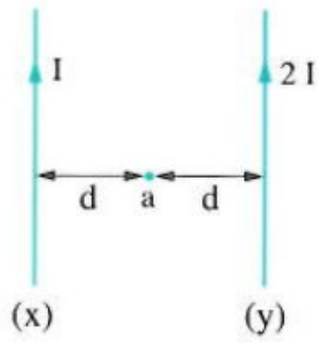
11 Two parallel straight wires (x, y) at 0.3 m apart, where wire x carries an electric current of intensity 2 A and wire y carries a current of intensity 3 A. **Calculate the distance between the neutral point and wire y, when the two currents are:**

(a) In the same direction. **(b)** In opposite directions.

12 Two parallel straight wires (1), (2), each of them carries an electric current as in the opposite figure. **Determine** the value of the resultant magnetic flux density at point C in terms of μ, d, I.

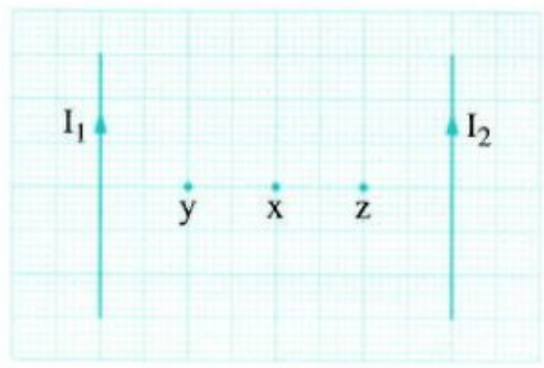


13 Two parallel straight wires (x, y), each of them carries an electric current as in the opposite figure. **Explain what happens to** the value of the magnetic flux density at point a when wire x gets moved a distance $\frac{1}{2}$ d towards wire y.

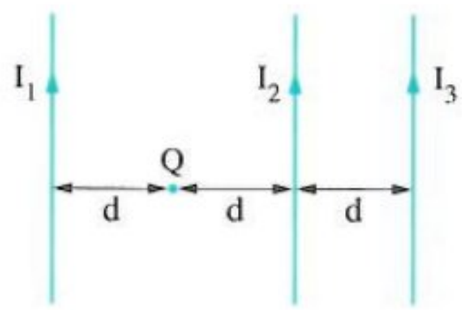


14 In the opposite figure, when the currents I_1, I_2 pass in the two long parallel straight wires, point x becomes neutral point. **How can the electric current intensity I_2 be changed to make the neutral point be:**

(a) at point y? **(b)** at point z?

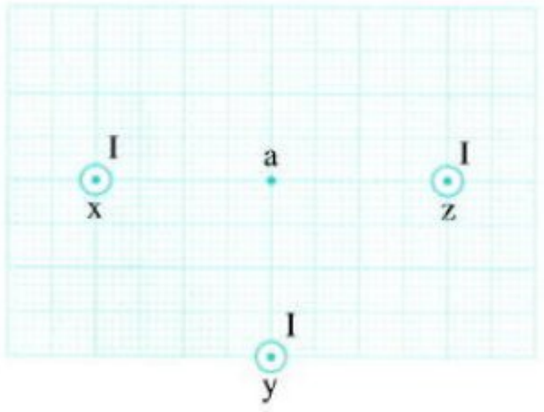


15 The opposite figure shows three long straight parallel wires that carry electric currents. If the net magnetic flux density at point Q was vanished, **prove that:** $(I_2 + I_3) > I_1$



16 In the opposite figure, three parallel straight wires x, y, z each of them carries an electric current of intensity I. If the magnetic flux density due to the current in wire x at point a equals B, **calculate** the value of the net magnetic flux density at point a in terms of B when the current vanishes in:

(a) wire y. **(b)** wire z.

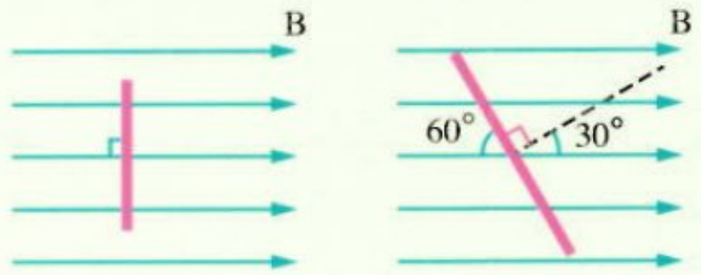


Scanned with OKEN Scanner

Questions of the Previous Years Exams ? on Lesson 1

The magnetic flux

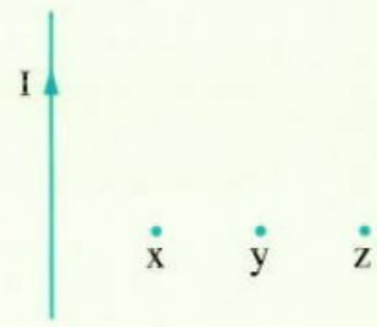
1 In the opposite diagram, two rectangular coils x and y, their areas respectively are A and 3A and are placed in a uniform magnetic field.
The ratio of the magnetic flux passing through each of them $\left(\frac{(\phi_m)_x}{(\phi_m)_y}\right)$ is ..... . *(Practice 25)*



(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{2\sqrt{3}}{9}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{2\sqrt{3}}{5}$

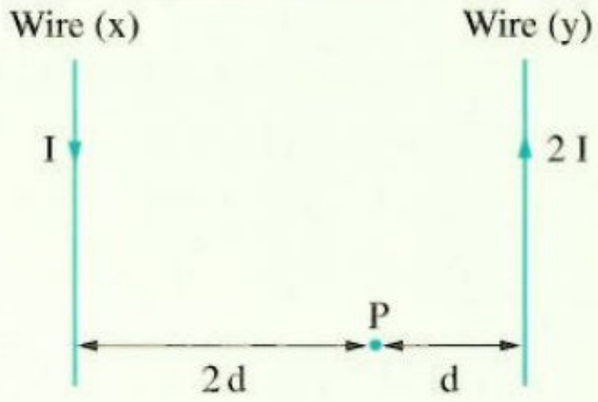
Straight wire

2 A long straight wire carries electric current of intensity I as in the figure, which one from the following expressions relates correctly the magnetic flux density (B) that is produced due to the current through the wire at the points x, y and z that lie in the same plane of the wire? *(Exp. June 21)*



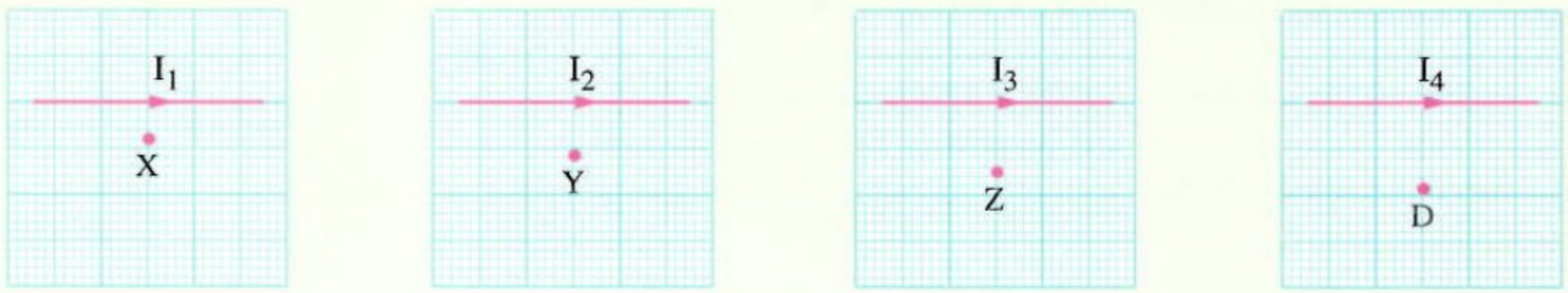
(a) $B_z > B_y$ (b) $B_y < B_x$ (c) $B_x < B_z$ (d) $B_y = B_z$

3 In the shown figure, the resultant magnetic flux density at point P due to the two electric currents that are passing through the two wires (x) and (y) is B_t, so if the direction of the passing current through wire (x) is reversed while keeping the current of (y) as it is, the resultant magnetic flux density at point P becomes ..... . *(Exp. June 21)*



(a) $\frac{3}{7} B_t$ (b) $\frac{2}{3} B_t$ (c) $\frac{3}{5} B_t$ (d) $\frac{3}{8} B_t$

4 The following figures show four straight wires that are carrying different currents (I_1, I_2, I_3 and I_4).



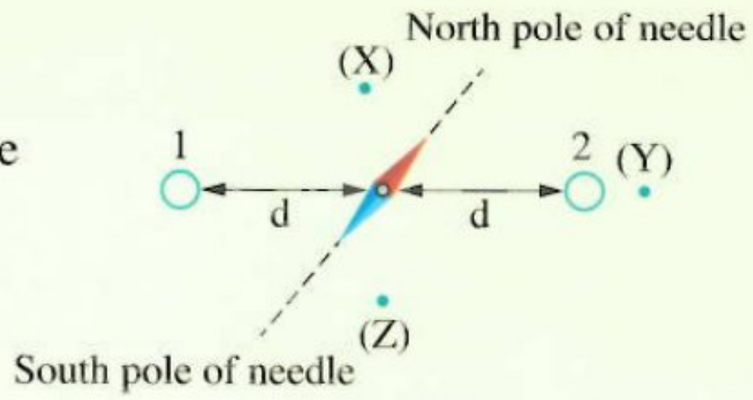
If the magnetic flux densities at the points (X, Y, Z and D) are equal, the greatest electric current intensity is ..... . *(1st session 21)*

(a) I_1 (b) I_2 (c) I_3 (d) I_4

CamScanner
Scanned with OKEN Scanner

CamScanner

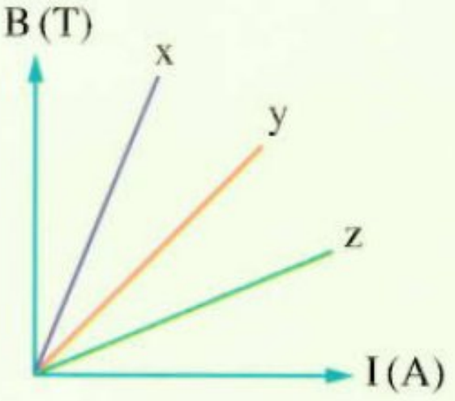
5 Two straight wires 1, 2 are in the same plane perpendicular to the plane of the page, a current of intensity I passes through each of them in the same direction, so if a magnetic needle is placed at the mid-point between them as in the figure, the north pole of the needle is expected to ..... .



(a) deflect till point X (b) deflect till point Y

(c) deflect till point Z (d) keep its orientation without deflection *(2nd session 21)*

6 The graph represents the relation between the magnetic flux density (B) at a point due to passing a current in the straight wire and the current intensity (I) for three straight wires (x, y and z), each separately so this point is ..... . *(2nd session 21)*



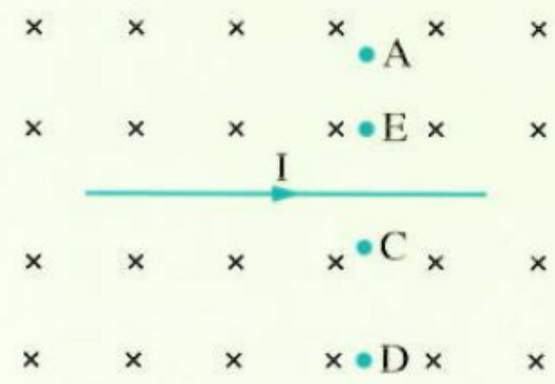
(a) nearer to wire (z) than wire (y)

(b) at equal distances from the wires (x, y, z)

(c) nearer to wire (x) than wire (y)

(d) nearer to wire (y) than wire (x)

7 The figure represents a straight wire carrying current (I) placed in a uniform magnetic field. So, the arrangement of the resultant magnetic flux density (B) at points (A, E, C, D) will be ..... . *(1st session 21)*



(a) $B_C > B_D > B_A > B_E$

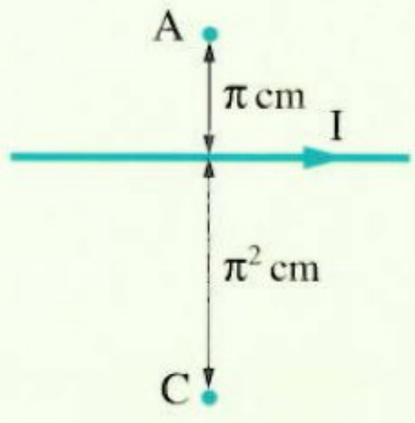
(b) $B_D > B_C > B_E > B_A$

(c) $B_A > B_C > B_D > B_E$

(d) $B_E > B_C > B_D > B_A$

8 A straight wire is carrying a current of intensity (I) as in the figure. A and C are two points beside the wire.

So, the ratio between: $\frac{\text{the magnetic flux density at A } (B_A)}{\text{the magnetic flux density at C } (B_C)} = ...$



(a) $\frac{1}{\pi}$ (b) $\frac{1}{2\pi}$

(c) 2π (d) π *(2nd session 22)*

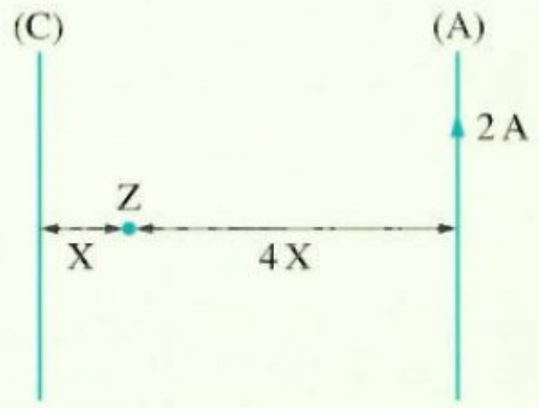
Scanned with OKEN Scanner

9 Two parallel long wires (X) and (Y), the normal distance between them is 0.5 m, the two wires are carrying currents in the same direction, the value of the current in wire X is (I) and that of wire Y is (3 I), so that the position of the neutral point is at a distance ..... .

(a) 0.125 m from wire Y (b) 0.25 m from wire Y

(c) 0.125 m from wire X (d) 0.625 m from wire X *(Exp. 23)*

10 The figure represents two long straight current carrying parallel wires (A) and (C). Which of the following choices represents the intensity and the direction of the current in wire (C) that produces a neutral point at point (Z)? *(1st session 23)*



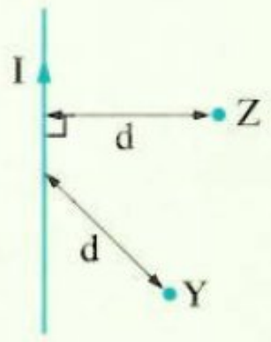
(a) 2 A, in the same direction of the current in wire (A).

(b) 0.5 A, in the same direction of the current in wire (A).

(c) 0.5 A, in opposite direction of the current in wire (A).

(d) 2 A, in opposite direction of the current in wire (A).

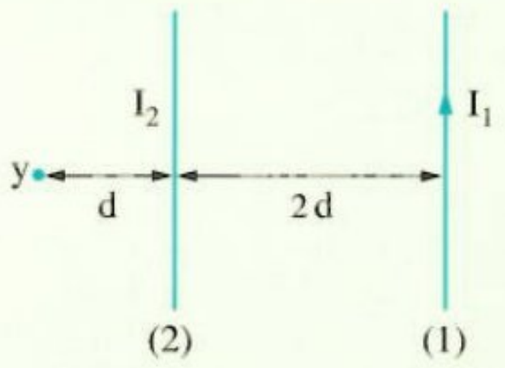
11 The figure illustrates a straight current-carrying wire. Which of the following choices represents correctly the relation between the magnetic flux densities (B_Y and B_Z) due to the current (I) in the wire at points (Y) and (Z) that lie in the same vertical plane with the wire? *(2nd session 23)*



(a) $B_Z = B_Y$ and in the opposite direction. (b) $B_Z = B_Y$ and in the same direction.

(c) $B_Z > B_Y$ and in the opposite direction. (d) $B_Z < B_Y$ and in the same direction.

12 The figure illustrates two parallel straight wires (1, 2) carrying currents (I_1, I_2) respectively. To make point (y) represent a neutral point for the two magnetic fields, the intensity and the direction of current I_2 must be ..... .



(a) $\frac{I_1}{2}$, upwards (b) I_1, upwards

(c) $2\ I_1$, downwards (d) $\frac{I_1}{3}$, downwards *(2nd session 23)*

13 When an electric current passes through a uniform cross-section straight wire of resistance (R) that is connected to a DC source and placed in air, a magnetic field of flux density (B) is generated at a point beside the wire. Which of the following changes leads to a decrease in the flux density at the same point, while keeping the other factors constant? *(1st session 24)*

(a) Replacing the wire with another one of smaller length.

(b) Replacing the wire with another one of greater length.

(c) Replacing the wire with another one of larger cross-sectional area.

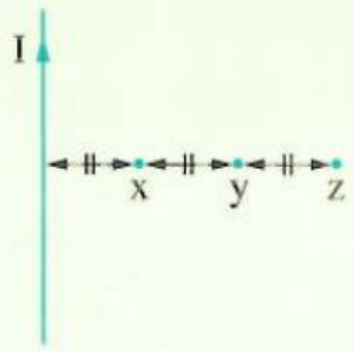
(d) Replacing the electric source with another one of higher electromotive force e.m.f

CamScanner

Scanned with OKEN Scanner

CamScanner

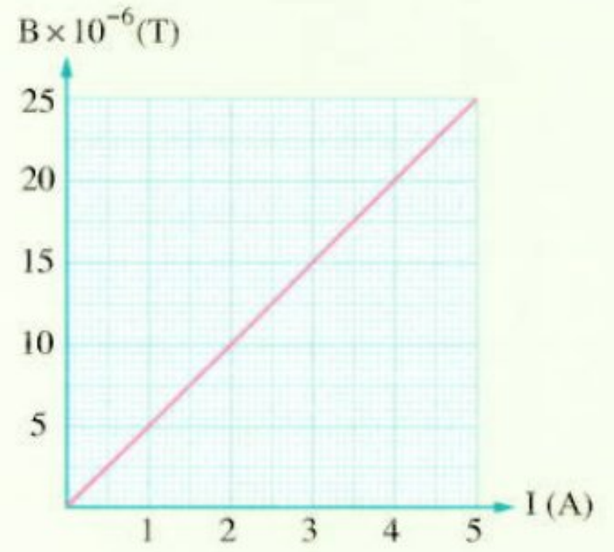
14 In the opposite figure, a straight wire carries an electric current, then the ratio between the magnetic flux density due to the current at the three points x, y, z ($B_z : B_y : B_x$) respectively is equal to ..... .



(a) 2 : 3 : 6 (b) 3 : 2 : 1

(c) 1 : 2 : 3 (d) 4 : 6 : 2 *(2nd session 24)*

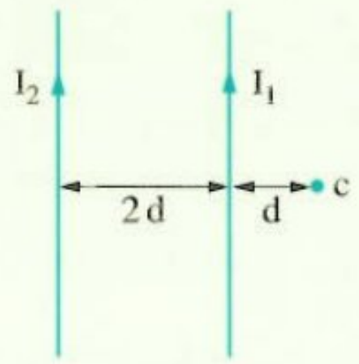
15 The opposite graph represents the relation between the magnetic flux density (B) generated by the current passing through a straight wire at a certain point and the intensity of this current (I). Therefore, the distance of that point from the axis of the wire equals ..... .

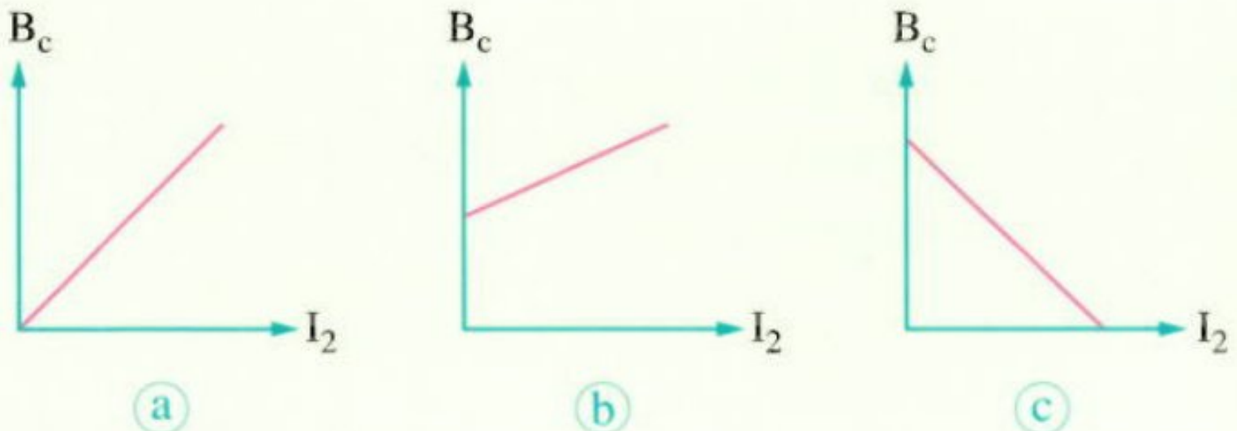


(a) 10 cm (b) 8 cm

(c) 6 cm (d) 4 cm *(Practice 25)*

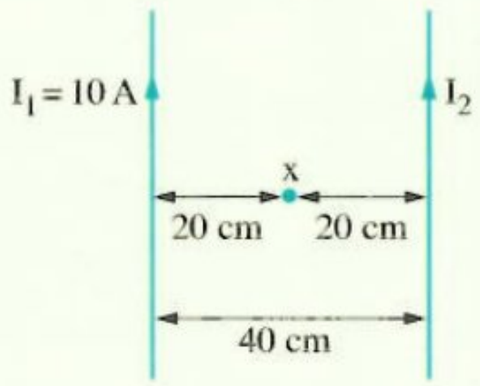
16 The opposite figure represents two long straight parallel wires carrying electric currents in the same direction, which of the following graphs represents the relation between the resultant magnetic flux density (B_c) of the two wires at point (c) and the current intensity (I_2)? *(Practice 25)*





(a) (b) (c) (d)

17 In the opposite figure, two very long straight parallel wires lie in the plane of the page, if the resultant magnetic flux density at point (x) generated by the currents of the two wires is 2×10^{-5} T, so the electric current intensity passing through the second wire (I_2) equals ..... .

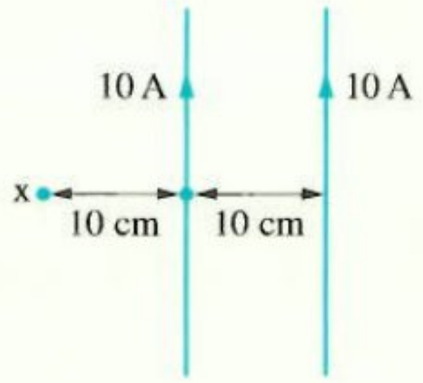


(Practice 25)

(a) 10 A (b) 20 A (c) 30 A (d) 40 A

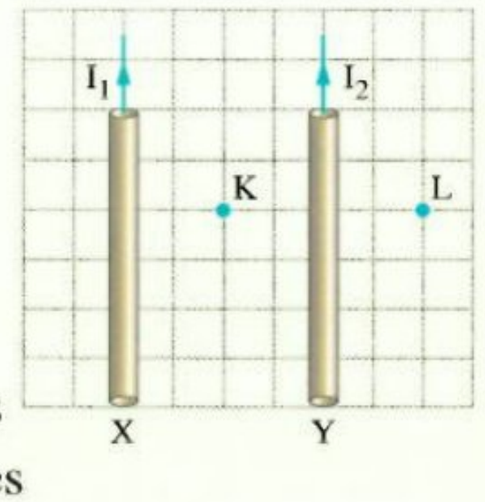
Scanned with OKEN Scanner

18 In the opposite figure, two very long straight parallel wires carry electric currents of the same intensity and are placed in the plane of the page, then the resultant magnetic flux density at point (x) equals ..... . *(Practice 25)*



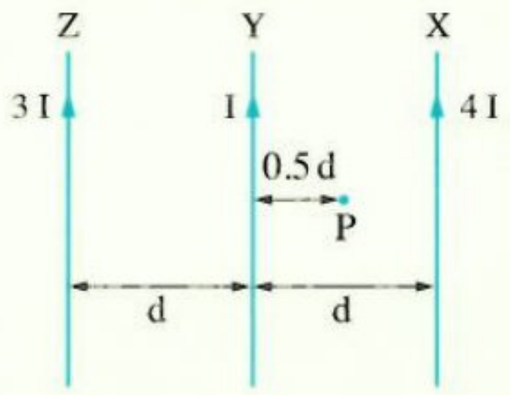
(a) 10^{-5} T
(b) 2×10^{-5} T
(c) 3×10^{-5} T
(d) 5×10^{-5} T

19 Two long straight parallel wires, each carrying an electric current as shown in the opposite figure. If the direction of the current in wire (x) is reversed, then the magnitudes of the net magnetic flux densities at points L and K, respectively, will be ..... . *(Practice 25)*



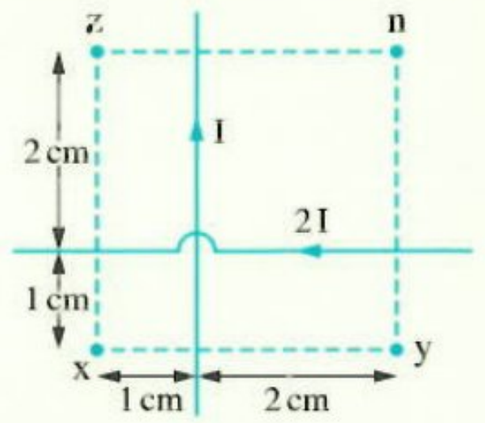
(a) decreases, increases
(b) increases, decreases
(c) unchanged, unchanged
(d) decreases, decreases

20 Three long straight parallel wires X, Y and Z, each carrying an electric current as shown in the opposite figure. If the magnetic flux density at point P due to the current in wire Y is equal to B, then the magnitude of the net magnetic flux density at point P is equal to ..... . *(Practice 25)*



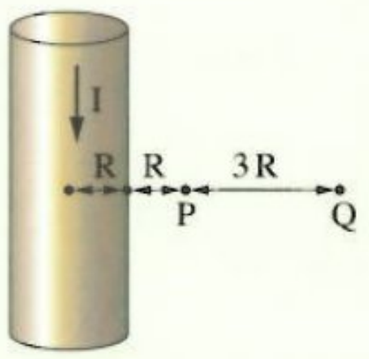
(a) B
(b) 2 B
(c) 4 B
(d) 8 B

21 The figure illustrates two straight, and insulated wires. They are located in the same plane and perpendicular to each other. An electric current passes through each wire.
Knowing that the four points (x, y, z, and n) shown in the figure are all in the plane of the two wires.
At which point of these four points, the net magnetic flux density is zero? *(2nd session 25)*



(a) Point (y).
(b) Point (x).
(c) Point (n).
(d) Point (z).

22 The figure shows a long straight wire with a uniform cross-sectional radius (R) through which a direct current (I) flows in the direction shown, so the relation between the magnetic flux density at point P and the magnetic flux density at point Q is ..... . *(Practice 26)*

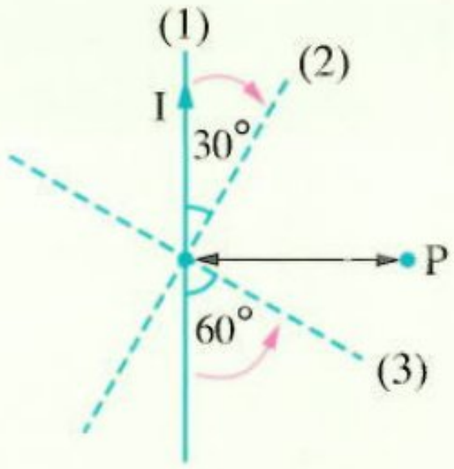


(a) $B_P = 2\ B_Q$
(b) $B_P = 3\ B_Q$
(c) $B_P = 4\ B_Q$
(d) $B_P = 2.5\ B_Q$

CamScanner
Scanned with OKEN Scanner

CamScanner

23 The opposite diagram represents a long straight wire carrying an electric current of intensity (I) and placed in position (1), so the resulting magnetic flux density at point P was (B_1). When the wire is rotated in the two directions shown in the diagram to become in position (2) and then in position (3), the magnetic flux densities at point P become (B_2) and (B_3), respectively. So, the correct arrangement for the magnitude of the magnetic flux densities at point P that are resulting from the passage of the same current in the wire in the three positions is ..... .



(Practice 26)

(a) $B_1 > B_2 > B_3$
(b) $B_3 > B_2 > B_1$
(c) $B_1 > B_3 > B_2$
(d) $B_2 > B_3 > B_1$

Scanned with OKEN Scanner